

AN EMPIRICAL FRAMEWORK FOR RAPID IBR SHORT CIRCUIT CALCULATIONS

by

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Abstract

The growing use of inverter-based resources (IBRs) introduces challenges for conventional short-circuit analysis because their fault-current response is governed by converter controls instead of machine impedance. This thesis investigates the short-circuit behavior of IBRs and develops a framework for rapid short-circuit current calculation based on high-resolution electromagnetic transient (EMT) simulations in PSCAD of a typical inverter. The effects of terminal-voltage magnitude and phase, prefault active and reactive power, phase-locked loop (PLL) behavior, and inverter current limitation are examined systematically. Based on these observations, a reduced-order empirical model is developed to estimate active power, reactive power, current magnitude, and current phase during faults. The model incorporates nonlinear voltage dependence, low-voltage controller-state freezing, and converter current limits. Validation against PSCAD simulations demonstrates that the proposed formulation can reproduce the principal fault-current characteristics while requiring significantly less model detail than a full EMT representation.

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Chapter 1

Introduction

1.1 Background

The rising deployment of renewable energy generation has led to an increasing proportion of electrical energy resources being connected to the power system through power-electronic converters instead of directly through synchronous machines (SMs). These IBRs include technologies such as solar photovoltaic generation, wind generation, and battery energy storage systems whose interaction with the network is determined mostly by their converter controls. As IBR penetration increases, the traditional synchronous-machine-dominated system is transitioning toward one in which voltage, frequency, and power exchange are more influenced by converter control algorithms. Unlike synchronous generators, whose electromagnetic coupling inherently establishes many aspects of their network response, IBRs commonly regulate injected current using P/Q control and, in the case of grid-following (GFL) converters, a phase-locked loop (PLL), while grid-forming resources regulate voltage and frequency through different control structures.

This transition from SMs to IBRs has important consequences during power system faults. The short-circuit contribution of a synchronous generator can be approximated from its internal voltage and short-circuit reactance (“ X'' ”), allowing it to be represented as a voltage source behind an impedance. However, the fault current of an IBR is not governed by an equivalent impedance; it depends on terminal voltage, converter-current capability, prefault operating conditions, synchronization behavior, current-priority logic, and fault-ride-through controls. Moreover, power-electronic devices have more restrictive temporary over-current capabilities than SMs, so the inverter controller must actively constrain the current when its limit is reached. Mahmud et al. [5] further show that different mechanisms dominate different portions of the fault response: the initial transient is determined by inverter filters and the severity of the voltage disturbance, whereas the current limiter becomes a dominant factor for the sustained fault current. Accurate fault-current modeling must represent how much current an IBR can supply and also how that current changes with the various factors in play.

These differences create significant challenges for transmission planning and protection studies, which have historically relied on relatively simple short-circuit representations of energy generating equipment. Cheng et al. [6] note that increasing IBR penetration is expected to fundamentally alter bulk-system short-circuit behavior and may require corresponding changes to conventional protection

settings and calculation methods. Protection functions can depend on both the magnitude and phase of fault current, and different IBR control strategies can produce different responses even for resources with similar ratings and connection impedance. IEEE Std. 2800 [7] further establishes fault-ride-through and current-injection requirements for transmission-connected IBRs, including behavior associated with abnormal voltage and sequence quantities, reinforcing the need for models capable of representing controller-dependent fault responses. Detailed EMT models can account for these effects with high fidelity, but their computational and model-information requirements make them difficult to apply routinely to large transmission systems.

1.2 Motivation

A central difficulty in IBR short-circuit analysis is the trade-off between model simplicity and representation. Conventional analytical models based on constant current, constant power, or simple current-limited sources can be computationally efficient, but they do not simultaneously preserve terminal-voltage dependence, prefault command P/Q dependence, current-priority behavior, and synchronization/current-phase effects. More detailed voltage-controlled current-source models can instead use manufacturer-specific lookup tables relating terminal voltage to fault-current magnitude and phase. However, Cheng et al. [6] note that constructing these tables requires knowledge of the inverter control settings and low-voltage ride-through logic, including P/Q priority, current limiting, and prefault I_d/I_q . When this characterization is not disclosed in manufacturer documentation, the response may have to be obtained by probing a detailed OEM model. Cheng et al. [6] also note that utilities sometimes resort to conventional generator representations with current limiters because appropriate inverter modeling information is unavailable. This difficulty is compounded by the fact that detailed vendor-specific models are commonly proprietary and may require OEM participation or nondisclosure agreements. Even when an accurate manufacturer-specific representation exists, it may not be accessible or generalizable for routine utility short-circuit studies involving many different IBR technologies.

Electromagnetic-transient (EMT) simulation provides a higher-fidelity alternative because it can explicitly represent the inner controllers and control logic of the inverter. However, this accuracy comes with steep computational and modeling requirements. EMT simulation solves the instantaneous three-phase network using small integration time steps and requires considerably more equipment and controller detail than conventional phasor-domain short-circuit calculations; its computational burden also increases with the network size and number of IBRs. NERC [8] recognizes detailed vendor-specific EMT models as important when simplified representations are inadequate, but such models are not necessarily practical as the routine calculation method for a large transmission network. This creates a practical need for models that retain the dominant fault-response characteristics of EMT simulations while remaining simple and fast for repeated planning and protection calculations.

The importance of this problem is illustrated by the 2016 Bluecut Fire event in Southern California. Multiple transmission-line faults occurred as the fire affected a major transmission corridor, and inverter controls at several bulk-system solar plants responded by ceasing current injection; the largest of such inverter shutdowns resulted in the loss of nearly 1,200 MW of solar generation, even though the facilities themselves were not directly disconnected by the transmission faults [9, 10].

The subsequent analysis identified inverter responses to abnormal voltage and erroneous frequency measurements as important contributors, demonstrating that system-level fault behavior can depend directly on inverter control logic and settings instead of only on network impedance [10]. The Bluecut Fire event provides a practical motivation for improved IBR fault modeling: transmission planners and protection engineers require models that are accurate enough to capture control-dependent behavior and computationally efficient to evaluate large numbers of resources, faults, and operating conditions. The objective of this thesis is to address this gap by using a detailed PSCAD EMT model as the benchmark from which the dominant relationships are identified and condensed into a faster, empirically derived model for rapid short-circuit calculations, with the understanding that future IBRs could be modeled using the same framework with comparable experimental measurements or detailed EMT simulation data for parameter identification.

1.3 Research Objectives

The proposed framework separately models the inverter’s postfault active power, reactive power, current magnitude, and current phase, while preserving the physical relationships among these quantities. The initial formulation begins from the positive-sequence dq -frame relationships, in which active and reactive power determine the d - and q -axis current components, their vector magnitude determines the total current, and their relative allocation determines the current orientation. Empirical corrections are introduced to account for the nonlinear voltage dependence, current limitation, and low-voltage controller-state behavior observed in the benchmark PSCAD simulations.

A second objective is to identify the dominant variables and control mechanisms governing IBR fault current, including terminal-voltage magnitude and phase, reference P and Q , prefault voltage, the converter current limit, current-priority logic, and the low-voltage freeze threshold. This study seeks to fit the EMT response and also to determine which variables must be retained in a reduced model to preserve the piecewise transitions between normal, current-limited, and low-voltage operating regions. Finally, the resulting empirical model is compared systematically with the PSCAD simulation data using errors in P , Q , current magnitude, and current phase, together with a 0.1-pu current-magnitude acceptance criterion. This validation is then extended beyond the original single-inverter cases to network-level studies to determine whether the reduced formulation remains accurate and numerically applicable when multiple IBRs interact through the surrounding power system.

1.4 Contributions Detailed in this Thesis

The primary contribution of this thesis is the development of a physics-informed empirical framework for rapid calculation of the sustained short-circuit response of IBRs, with demonstrations based on a detailed, open-source GFL inverter model published by EPRI [4]. The framework is designed to occupy an intermediate position between highly simplified constant-current or constant-power representations and computationally intensive EMT simulations. Instead of assuming that the inverter maintains $P = P_{\text{ref}}$ and $Q = Q_{\text{ref}}$ throughout a disturbance, the proposed approach retains the dominant nonlinear dependencies identified from exhaustive PSCAD simulations.

A second contribution is the development of separate empirical formulas for the inverter’s active power, reactive power, current magnitude, and current phase during the sustained fault period. The active- and reactive-power models combine fundamental dq -frame relationships with empirical corrections derived from the observed low-voltage PSCAD response. Current magnitude is then reconstructed from the calculated P and Q using the physical complex-power relationship and constrained by $I_{\max}$. A corresponding phase formulation preserves the dominant dependence of current orientation on the P/Q allocation, allowing the inverter fault contribution to be represented as a phasor. This structure maintains physical consistency among P , Q , and I while keeping the number of empirical parameters relatively small.

The thesis also contributes a detailed physical investigation of the variables and controller mechanisms governing IBR fault behavior. In particular, the work distinguishes the normal, current-limited, and low-voltage “frozen” (defined below) operating regions and demonstrates why a single linear or constant-current relationship cannot describe the complete fault response. The study also examines current phase explicitly and shows that reproducing current magnitude alone is insufficient because the current orientation depends on both the d/q -axis current allocation and synchronization state, with especially nonlinear behavior near V_{freeze} and at very low voltage.

Finally, the thesis outlines a progressive validation and scalability assessment of the proposed framework. The proposed formulations are first compared directly with the PSCAD EMT dataset, after which the method is extended to an interconnected system containing multiple IBRs with different prefault operating points. The model is then embedded in an iterative network power-flow formulation so that inverter injections and terminal voltages are solved self-consistently instead of treating each resource independently. A final convergence study using a synthetic 4000-bus system demonstrates that the nonlinear empirical formulation can be incorporated into a large sparse network simulation. Together, these contributions establish a framework that retains the principal control-dependent features of the sustained IBR short-circuit response while remaining sufficiently compact for rapid and large-scale power-system studies.

Chapter 2

Literature Review

2.1 Inverter-Based Resources

Inverter-based resources (IBRs) represent electricity generation technologies in which the electrical interface between the energy source and the power system is provided through power-electronic converters. The increasing deployment of converter-interfaced resources has introduced control-based methods for regulating voltage, frequency, and power exchange with the grid.

2.1.1 Definition and Historical Development of Inverter-Based Resources

The Consortium for Electric Reliability Technology Solutions (CERTS) Microgrid was conceived as a collection of distributed energy resources and loads capable of operating connected to the utility grid or autonomously in isolation while maintaining stable voltage and frequency without requiring centralized supervision [11]. Unlike conventional power systems that rely on synchronous generators (SGs) to establish the grid voltage and frequency, the CERTS Microgrid demonstrated that power electronic converters could perform many of the same functions as an SG with appropriate control systems. It primarily relies on a decentralized architecture with distributed micro-sources, local loads, and a switch to separate the microgrid from the main grid during faults, with each inverter using only locally measured electrical quantities for its operations [11]. This technique allows IBRs to be integrated without redesigning the overall control system of the grid.

A major contribution of [11] is the introduction of droop-based control for IBRs. Similar to SGs, active power is regulated using a power-frequency ($P-\omega$) droop relationship, while reactive power is regulated using a voltage-reactive power ($Q-V$) characteristic [11]. When operating in an isolated microgrid, an increase in load causes the system frequency to decrease, prompting each inverter to proportionally increase its active power output. Likewise, reactive power adjustments regulate the inverter voltage, preventing circulating reactive currents between multiple distributed generators.

While “grid-forming inverters” was not commonly used as a term at the time of [11]’s publication, the autonomous voltage and frequency control demonstrated by the CERTS controllers forms the foundation of modern grid-forming control strategies. Modern grid codes require current-limited fault ride-through, positive- and negative-sequence current injection, and detailed transient current behavior specified in standards such as IEEE Std. 2800 [7].

2.1.2 Grid-Following and Grid-Forming Inverter Control

Geng and Hiskens [12] examine the distinction between grid-following (GFL) and grid-forming (GFM) inverter control. Conventional GFL inverters use a PLL to synchronize their internal reference frame to the voltage at the terminal point of connection (POC). This approach requires an already established grid voltage that is led by other SGs or GFM inverters and can become problematic in weak grids when variations in the inverter output significantly disturb the voltage the PLL uses for synchronization. In contrast, GFM control does not fundamentally depend on direct locking to an established voltage. GFM inverters use an active- $P - \omega$ droop relationship to participate in forming the grid instead of synchronizing to the grid's frequency [12]. This distinction allows the GFM inverters to energize a network from a black-start.

2.1.3 Equivalent Network Representations of GFL and GFM Inverters

The fundamental difference between GFL and GFM control can also be described using simplified network equivalents. These representations remove the detailed internal controller structure and instead characterize the electrical behavior of the inverter as seen by the surrounding power system. Singh et al. [2] explicitly describe GFL inverters as converter-controlled current sources. On the other hand, the primary objective of the GFM internal loop is to regulate output voltage, allowing the GFM inverter to be represented as a converter-controlled voltage source.

A GFL inverter can be represented as a controlled current source connected to the point of connection (POC) through the inverter filter and transformer impedance. The surrounding grid establishes the terminal-voltage magnitude, phase, and frequency, while the inverter controller determines the magnitude and phase of the injected current according to the active- and reactive-power references. By contrast, the GFM inverter is represented as a converter-controlled internal voltage source behind an equivalent impedance. Its output current is not imposed directly as the primary controlled quantity, but instead results from the difference between the internally established inverter voltage and the network voltage.

For the GFL representation, the injected current is determined from the desired power exchange and the measured POC voltage. Singh et al. [2] note that GFL resources receive active- and reactive-power references and use measured voltage and current quantities at the point of interconnection to determine the current that must be injected into the network. In phasor form, the complex power output of the inverter is

$$\underline{S}_{\text{POC}} = P + jQ = \underline{V}_{\text{POC}} \underline{I}_{\text{GFL}}^* \quad (2.1)$$

where $\underline{V}_{\text{POC}}$ is the voltage at the point of connection and $\underline{I}_{\text{GFL}}$ is the current output of the inverter measured from the POC. The current corresponding to a specified active- and reactive-power reference can be written as

$$\underline{I}_{\text{GFL}} = \left(\frac{P + jQ}{\underline{V}_{\text{POC}}} \right)^* \quad (2.2)$$

Equation (2.2) illustrates the current-source interpretation of GFL control. For a given P and Q , the inverter modifies its current according to the measured terminal voltage. The complete controller contains additional dynamics, including the PLL and inner current-control loops, but the controlled-current-source representation provides a useful approximation of the inverter's fundamental interaction with the external network. Singh et al. [2] identify the inner current-control loop

and PLL as key components of generic GFL representations used in system-level studies.

Ravanji et al. [3] further demonstrate that GFL power exchange is coupled to the grid impedance and voltage angle instead of being completely independent of network conditions. For their grid-connected GFL model, assuming the voltage phase at the POC is 0° , the steady-state POC equations are

$$V_{\text{INV}} = V_{\text{POC}} \cos \delta + R_l I_d - X_l I_q \quad (2.3)$$

and

$$0 = -V_{\text{POC}} \sin \delta + R_l I_q + X_l I_d \quad (2.4)$$

where $V_{\text{INV}} \angle \delta$ is the inverter terminal voltage phasor, R_l/X_l the equivalent resistance and reactance between the POC and the inverter, and I_d/I_q the dq components of the inverter output current. The resulting active and reactive powers at the POC are

$$P = V_{\text{POC}} I_{\text{GFL}} \cos(\delta + \theta_I) + R_l I_{\text{GFL}}^2 \quad (2.5)$$

and

$$Q = V_{\text{POC}} I_{\text{GFL}} \sin(\delta + \theta_I) + X_l I_{\text{GFL}}^2 \quad (2.6)$$

where θ_I is the relative current phase to the inverter terminal voltage. These expressions show that although a GFL inverter is controlled as a current source, the actual P and Q delivered at the POC are influenced by the current magnitude and phase, grid impedance, and phase difference between the inverter and the grid. This becomes more pronounced in weak networks, where the voltage drop across the network impedance is not negligible.

The GFM equivalent is fundamentally different. Because the inverter regulates an output voltage, its internal behavior can be approximated as a controlled voltage source $\underline{E} = E \angle \delta$ connected to the network through an equivalent impedance $\underline{Z}_{\text{eq}} = R + jX$. The resulting inverter current is determined by the voltage difference across this impedance:

$$\underline{I}_{\text{GFM}} = \frac{\underline{E} - \underline{V}_{\text{POC}}}{\underline{Z}_{\text{eq}}} \quad (2.7)$$

Singh et al. [2] note that GFM implementations can regulate voltage either directly through the modulation command or indirectly through cascaded voltage and current control loops; both approaches retain the fundamental voltage-source behavior.

The complex power delivered by the GFM inverter can be written as $\underline{S} = \underline{V}_{\text{POC}} \underline{I}_{\text{GFM}}^*$ or, using [3],

$$\underline{S} = \underline{V}_{\text{POC}} \left(\frac{\underline{E} - \underline{V}_{\text{POC}}}{\underline{Z}_{\text{eq}}} \right)^* \quad (2.8)$$

Ravanji et al. [3] derive a corresponding steady-state formulation for a GFM inverter connected to a grid through an impedance $Z_{\text{eq}} = Z_l \angle \theta_l$. The active and reactive powers at the POC are expressed as

$$P = -\frac{V_{\text{POC}} E}{Z_l} \cos(\delta + \theta_l) + \frac{V_{\text{POC}}^2}{Z_l} \cos \theta_l \quad (2.9)$$

and

$$Q = -\frac{V_{\text{POC}}E}{Z_l} \sin(\delta + \theta_l) + \frac{V_{\text{POC}}^2}{Z_l} \sin \theta_l, \quad (2.10)$$

assuming the POC voltage phase angle is 0° . These relationships explicitly connect GFM active and reactive power to the POC voltage, grid voltage, impedance magnitude and angle, and the voltage phase displacement δ . For a predominantly inductive equivalent impedance, $R \ll X$, these relationships can be interpreted using the familiar approximate power-angle behavior of a voltage source behind reactance (where $\theta_l = \frac{\pi}{2}$):

$$P \approx \frac{EV_{\text{POC}}}{X} \sin \delta \quad (2.11)$$

while reactive-power transfer is principally determined by the voltage-magnitude difference and power angle,

$$Q \approx \frac{V_{\text{POC}}^2 - EV_{\text{POC}} \cos \delta}{X} \quad (2.12)$$

where the sign of Q depends on whether positive reactive power is defined as injection into or absorption from the network. Geng and Hiskens [12] relate inverter active power to the phase difference across the inverter filter inductance and show that regulating this phase difference directly controls active-power output.

The current-source and voltage-source equivalents should nevertheless be interpreted as system-level approximations instead of ideal physical sources. A practical GFL inverter cannot produce arbitrary current because its semiconductor devices impose a finite current capability, while a practical GFM inverter cannot maintain an arbitrary voltage behind a disturbance when its current limit is reached. In addition, physical filter impedance, transformer impedance, virtual impedance, inner control loops, and voltage limits modify the actual response, especially during faults and weak-grid operation.

2.1.4 Converter-Driven Stability

Singh et al. [2] investigated slow-interaction converter-driven stability (SICDS) in medium-voltage systems containing IBRs. The paper is motivated by the emergence of IBR-specific stability phenomena resulting from the increasing penetration of power-electronic generation. SICDS arises from dynamic interactions between the inverter control systems and other components of the power grid. They distinguish slow interactions from fast converter-driven phenomena by frequency: fast-interaction issues are typically associated with high-frequency resonances involving LCL filters and inner current-control loops, whereas SICDS includes oscillatory phenomena below 50 Hz and can arise through interactions between geographically separated IBRs. This work explains the oscillatory behavior of the inverters in Section 6.2.

Singh et al. argue that an inverter may be appropriately tuned from the perspective of an individual plant owner and still participate in unstable interactions once connected to a network containing other IBRs [2]. Device-level tuning alone is not sufficient to guarantee system-wide stability. Unlike fast converter-driven interactions that can often be addressed through local filter design or controller damping, SICDS may require coordinated, system-level measures. In this case, the growing number of decentralized IBRs makes detailed vendor-specific modeling increasingly impractical. Singh et al. emphasize the use of generic, vendor-agnostic inverter models that preserve the control blocks most

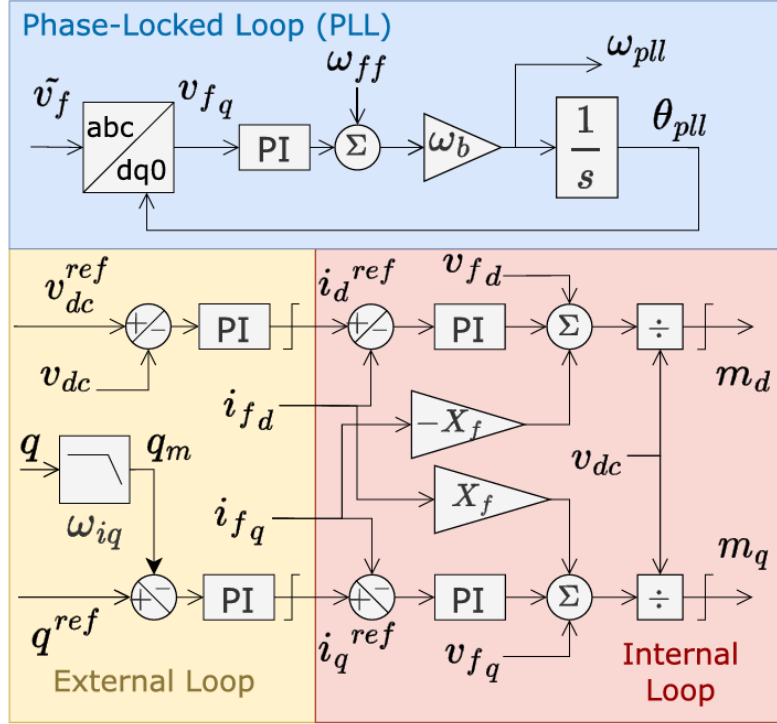


Figure 2.1: Control Diagram for a Grid-Following Inverter as Presented in [2].

relevant to low-frequency dynamic behavior [2]. For GFL inverters, the authors identify the PLL and current-control structure as fundamental components for reproducing converter-driven stability behavior. They note that generic models can be parameterized to approximate the behavior of a broad range of actual implementations, providing utility companies with a practical basis for system-level planning and stability assessment [2].

The difference between GFL and GFM control is essential to the analysis. The local dq reference frame of a GFL inverter is obtained from a PLL synchronized to the terminal voltage as in [12] with a similar control hierarchy (Fig. 2.1). For the GFM inverter, its internal control regulates terminal voltage, and its synchronization mechanism is based on active-power control and droop (Fig. 2.2). When multiple GFM inverters operate together, droop-based control allows power sharing similar to SGs. Singh et al. [2] also discuss virtual-synchronous-machine interpretations, in which a low-pass filter combined with $P - \omega$ droop allows the inverter to emulate the slower inertial response of an SM.

The core of Singh et al.'s methodology is a small-signal stability analysis of multiple interconnected IBRs [2]. The results show that stability depends on the interaction between controller parameters and the surrounding network in addition to the type of IBRs [2]. Just like PLL-related dynamics are important for GFL resources, $P - \omega$ droop and filtering parameters are important for GFM resources. This shows that converter-driven instability is a system interaction problem rather than a simple consequence of a poorly designed inverter.

Singh et al. ultimately propose a methodology in which a distribution system operator (DSO) first identifies controller parameters that strongly influence system eigenvalues, establishes acceptable parameter ranges, tests corner cases, and then uses those bounds to maintain stability [2]. Relatively

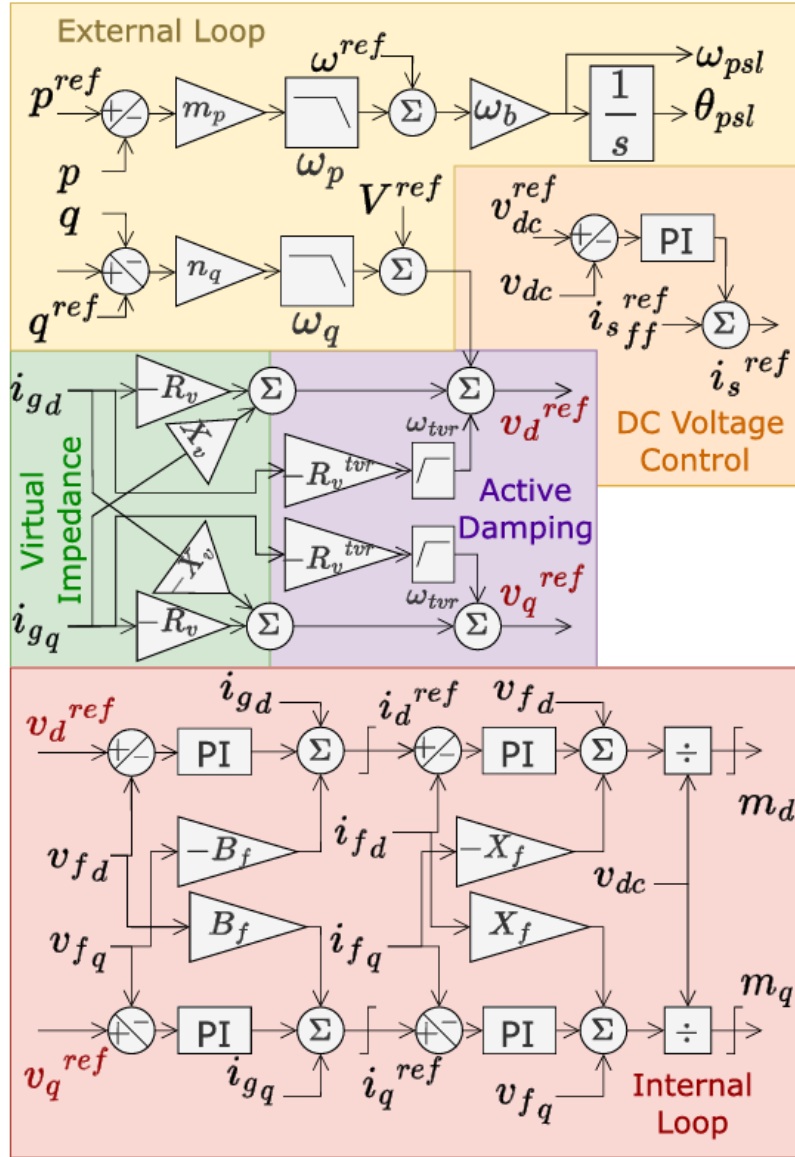


Figure 2.2: Control Diagram for a Grid-Forming Inverter as Presented in [2].

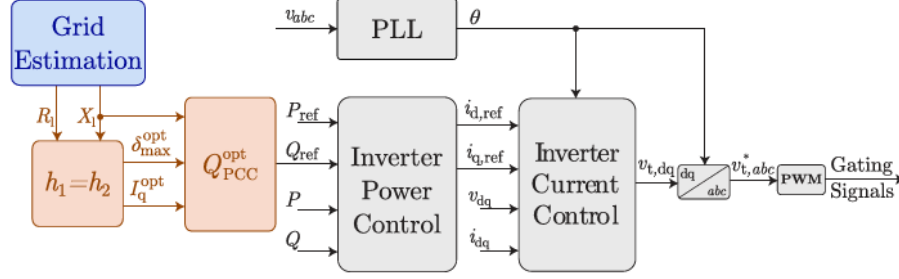


Figure 2.3: Auxiliary Control Loop for Maximizing the GFL Maximum Transferable Power while its Thermal Limits are not Exceeded [3]. The diagram summarizes the main control loops of the GFL inverter.

simple limits on a small number of generic parameters can potentially mitigate interactions between physically distant IBRs without requiring the DSO to know the proprietary details of each inverter controller. As IBR penetration increases, utilities cannot assess resources independently and assume that individually stable plants will remain stable after interconnection. Instead, DSOs can use generic models and standardized requirements that explicitly account for interactions among multiple inverter controls.

2.1.5 Inverter Current Constraint and Voltage-Dependent P - Q Capability

The PLL introduces an additional limitation on the maximum allowable power angle associated with PLL stability. This limit depends on PLL gains and network conditions. Even if the static network equations and inverter current rating allow a particular operating point, the GFL inverter can lose synchronization before reaching that point if the PLL is not tuned properly [3]. With a well-tuned PLL, the static or thermal constraint may determine the maximum power that can be transferred. However, PLL tuning itself restricts the PLL-induced boundary and reduces the achievable $P - Q$ operating region. Under unfavorable conditions, an operating point that satisfies the static and thermal constraints will become inaccessible due to PLL instability.

Based on the interaction between reactive-current support and thermal current limitation, Ravanji et al. propose an auxiliary controller for maximizing GFL active-power transfer (Fig. 2.3). An optimum I_q is found where the maximum static power-angle boundary intersects the thermal boundary [3]. Grid-impedance information is estimated and used to calculate the reactive-power reference corresponding to this optimum operating condition. The controller intentionally adjusts reactive power to increase active-power transfer while preserving the inverter current limit. Assuming a $\delta_{\max}^{\text{opt}}$ and a I_q^{opt} that maximizes the transferable power, using Equation (2.6) as a basis, the resulting Q_{opt} at POC will be

$$Q_{\text{POC}}^{\text{opt}} = -V_{\text{POC}} I_{\text{GFL}} \sin \left(\delta_{\max}^{\text{opt}} + \tan^{-1} \left(\frac{I_q^{\text{opt}}}{\sqrt{I_d^2 - (I_q^{\text{opt}})^2}} \right) \right) + X_l I_{\text{GFL}}^2 \quad (2.13)$$

The treatment of GFM inverters differs because their POC voltage is regulated directly. Equations (2.9) and (2.10) explicitly relate P and Q to POC voltage, network impedance, and power angle [3].

An important difference between the two control modes is their reactive-power behavior. As GFM active-power output increases, the controller naturally modifies reactive-power output in order to maintain the referenced POC voltage. A conventional GFL inverter operating at a fixed reactive-current or reactive-power reference does not automatically provide the same response [3]. Its POC voltage can decrease as active-power transfer increases. Thus, the GFL inverter is more susceptible to reaching its static or PLL-induced limits in weak networks unless additional reactive-power support is included.

To summarize, Ravanji et al. conclude that GFM inverters provide a larger maximum transferable power capability than conventional GFL inverters under the same conditions because they are not constrained by the PLL-induced stability boundary [3]. GFL capability is strongly affected by grid strength, PLL tuning, reactive-current reference, and converter current limits, whereas GFM capability is principally shaped by static network constraints, voltage regulations, and current limits. This paper shows that inverter P , Q , voltage, current, phase angle, and grid impedance can be connected through relatively compact algebraic relationships. At the same time, it demonstrates why an analytical model should not simply impose $P = P_{ref}$ and $Q = Q_{ref}$ under all operating conditions. The feasible output must respect a current constraint such as

$$I_d^2 + I_q^2 \leq I_{\max}^2 \quad (2.14)$$

while the resulting active and reactive power depend on terminal voltage, grid strength, control mode, and synchronization behavior.

2.1.6 Comparison of GFL and GFM Inverters

The following table summarizes the characteristics of GFL & GFM converters as discussed above:

Table 2.1: Comparison of grid-following and grid-forming inverter characteristics.

Characteristic	Grid-Following Inverter	Grid-Forming Inverter
Network representation	Controlled current source	Controlled voltage source behind impedance
Synchronization	Typically PLL-based	Internally established angle/frequency
Main grid dependence	Requires an established voltage waveform	Can establish voltage and frequency
Typical active-power mechanism	I_d / active-power controller	Power angle and $P-\omega$ droop
Typical reactive-power mechanism	I_q / reactive-power controller	Voltage regulation and $Q-V$ droop
Primary regulated quantities	P, Q or I_d, I_q	Voltage and frequency
Weak-grid sensitivity	Strongly affected by PLL/network interaction	No conventional PLL-induced limit
Current limitation	Required	Required
Fault-current response	Controller dependent	Controller dependent
Black-start capability	Generally not inherent	Can be provided

2.2 Short-Circuit Behavior

The short-circuit behavior of IBRs differs fundamentally from that of conventional SGs. In an SG, fault current is principally determined by the machine's internal electromotive force and its electromagnetic reactances. Conventional short-circuit calculations can approximate the machine as a voltage source behind an impedance. In an IBR, the current delivered during a disturbance is strongly influenced by the converter controller, current limit, voltage measurement, synchronization system, and fault-ride-through logic. Two IBRs with comparable power ratings and network impedances can provide different fault-current responses if their control strategies or current-priority schemes differ.

2.2.1 Difference Between Synchronous-Machine and IBR Fault Current

A simplified synchronous-generator fault contribution may be represented by

$$\underline{I}_{\text{SG}} \approx \frac{E'' - V_f}{jX_d'' + Z_{\text{net}}} \quad (2.15)$$

where E'' is the internal subtransient voltage, X_d'' is the subtransient reactance, V_f is the faulted-bus voltage, and Z_{net} represents the impedance of the interconnection network [13]. This representation illustrates the primarily impedance-determined nature of the initial contribution of the SM fault. An IBR does not possess the same electromagnetic relationship between internal flux, machine reactance, and fault current. Instead, the current supplied by the converter is generated through switching devices under the supervision of its control system. A general IBR fault-current representation is better expressed as

$$\underline{I}_{\text{IBR}} = f(\underline{V}, P_{\text{ref}}, Q_{\text{ref}}, \theta_{\text{PLL}}, I_{\text{max}}, \text{control mode, current priority, prefault state}) \quad (2.16)$$

where the function $f(\circ)$ represents the converter control and limiting logic. The terminal-voltage magnitude and phase, prefault operating point, synchronization state, and converter current capability can all influence the measured fault contribution.

For a GFL inverter, the controller determines the d - and q -axis current references based on voltage measurements and power-control objectives. The resulting current must remain within the current capability shown in Equation (2.14). Thus, once a voltage disturbance causes the unconstrained current reference to exceed this boundary, the converter control must modify one or both current components. The resulting active and reactive powers depart from their original references. The finite current capability established previously in Section 2.1.5 is one of the principal differences between IBRs and conventional SGs. The distinction is also applicable to GFM inverters. Although a GFM inverter is represented in the positive-sequence network as a controlled voltage source, the converter cannot maintain the referenced internal or terminal voltage indefinitely. Doing so would require a current greater than its physical capability. Nasr and Hooshyar [14] explicitly note that a current-limited GFM inverter cannot maintain its terminal voltage at 1 pu for all low-voltage ride-through conditions because severe faults would require a large current through the low-impedance fault path. The voltage-source representation of a GFM inverter becomes constrained by converter current capability during severe disturbances.

For GFL inverters, Mahmud et al. [5] show that the inverter filter and the severity of the terminal-voltage disturbance strongly influence the initial transient current spike, whereas the current limiter becomes the dominant factor governing the steady-state fault-current magnitude (This effect can be observed in Fig. 3.8). Nasr and Hooshyar [14] similarly show that GFM current-limiting mechanisms may not act instantaneously. In their proposed controller, virtual impedance successfully constrains the positive-sequence steady-state fault current, but the finite bandwidth of the voltage-control loop necessitates an additional transient suppressor to limit current during the initial portion of the disturbance. Since the framework proposed by this thesis focuses on GFL short-circuit behavior, the intricacies of the GFM short-circuit behavior will not be examined further.

The phase of the fault current represents another important difference. For an SG, the fault-current angle is predominantly established by the machine and network impedance together with the internal generator voltage. In an IBR, the current phase can additionally be altered by the converter's control objectives, synchronization systems, current-priority logic, and sequence-control strategies. Lyu et al. [15] emphasize that IBR fault characteristics can vary between manufacturers because they depend on control algorithms and settings, making their behavior more difficult to generalize using conventional source models. Since protection functions depend on both the magnitude and angle of the measured current, these characteristics create difficulties for conventional phasor-domain short-circuit programs.

2.2.2 Voltage Depression and Current-Limited Operation

During a grid fault, the reduction in terminal voltage can force an IBR away from its normal active- and reactive-power operating point. Under normal conditions, the outer control loops of a GFL inverter can regulate active and reactive power by generating corresponding d - and q -axis current references. When the synchronous reference frame is aligned with the terminal-voltage vector such that $V_q \approx 0$ and $V_d = V$, the power equations can be approximated in per unit as

$$P \approx VI_d \quad (2.17)$$

and

$$Q \approx -VI_q \quad (2.18)$$

where the sign of I_q assumes that Q is the *output* reactive power from the inverter. These relationships provide a direct connection between terminal-voltage depression and the current demanded from the converter. If the controller attempts to maintain the prefault active- and reactive-power references during a voltage disturbance, the corresponding unconstrained current references are approximately

$$I_d = \frac{P_{\text{ref}}}{V} \quad I_q = -\frac{Q_{\text{ref}}}{V}. \quad (2.19)$$

The magnitude of the requested inverter current is

$$I_{\text{req}} = \sqrt{(I_d)^2 + (I_q)^2} = \frac{\sqrt{P_{\text{ref}}^2 + Q_{\text{ref}}^2}}{V}. \quad (2.20)$$

Equation (2.20) demonstrates that maintaining constant reference P and Q requires increasing current as the terminal voltage decreases. This relationship cannot continue indefinitely because the semiconductor switches and associated converter hardware have a finite current capability as established in Section 2.1.5. The allowable current vector can be represented by the circular constraint in Equation (2.14). Combining Equations (2.17), (2.18), and (2.14) gives

$$P^2 + Q^2 \leq V^2 I_{\max}^2. \quad (2.21)$$

Equation (2.21) has an important physical implication during a fault: the maximum apparent power that can be transferred by a current-limited inverter decreases with terminal voltage. Even if P_{ref} and Q_{ref} remain unchanged, the feasible $P - Q$ operating region contracts as the voltage falls.

Mahamedi et al. [16] present an explicit fault-ride-through current-limiting strategy for IBRs. Their sequence-based controller separates positive- and negative-sequence quantities during unbalanced disturbances and applies a limiter directly to the phase currents. A crucial feature of their approach is that the limiter operates in the natural reference frame and constrains each physical phase to a predefined current threshold from the beginning of the disturbance. Likewise, Mahmud et al. [5] propose a short-circuit model that separates the initial transient behavior from the later current-limited response. This distinction is important because the current measured immediately following fault inception may differ significantly from that measured after the controller has entered its sustained current-limited state.

2.2.3 GFL Fault Response: PLL and dq Current Control

For a GFL inverter, the fault-current response is closely coupled to the operation of the PLL and the dq -axis current-control system. A conventional GFL uses the PLL to estimate the phase of the voltage measured at the POC, and this estimated angle defines the local synchronous reference frame used by the outer power-control and inner current-control loops. Singh et al. [2] identify the PLL, external control loop, and inner current-control loop as the principal components of a generic GFL controller, as illustrated in Fig. 2.1. Under normal balanced operation, the PLL tracks the terminal-voltage angle such that $\theta_{PLL} \approx \theta_V$ and the local d -axis is approximately aligned with the terminal-voltage phasor with magnitude V and angle θ_V :

$$V_d \approx V, \quad V_q \approx 0 \quad (2.22)$$

allowing active and reactive power to be regulated primarily through the d - and q -axis current components, respectively.

During a fault, the terminal-voltage magnitude and phase can change rapidly while the PLL has finite dynamics. The instantaneous synchronization error may be defined as

$$\Delta\theta = \theta_V - \theta_{PLL} \quad (2.23)$$

Under normal synchronized operation, $\Delta\theta \approx 0$, but immediately following a disturbance the PLL reference may temporarily differ from the actual terminal-voltage angle. Geng and Hiskens [12] describe the PLL as dynamically adjusting its internal phase to drive this phase difference toward zero; the estimated PLL angle provides the local reference used throughout the inverter controller.

The importance of this phase difference becomes apparent when the terminal voltage is expressed in the PLL-oriented reference frame. Its dq components can be written as

$$V_d = V \cos \Delta\theta \quad V_q = V \sin \Delta\theta \quad (2.24)$$

Thus, when $\Delta\theta = 0$ the condition in Equation (2.22) is recovered. During a fault-induced phase difference, $V_q \neq 0$ and the simplified decoupling between active and reactive power is inaccurate. Using the per-unit dq -frame power relationships

$$P = V [I_d \cos \Delta\theta + I_q \sin \Delta\theta] \quad Q = V [I_d \sin \Delta\theta - I_q \cos \Delta\theta] \quad (2.25)$$

Equation (2.25) shows that a fault-induced PLL angle error couples the d- and q-axis currents into both active and reactive power. Terminal-voltage phase and PLL state can affect the measured P and Q even when the current components themselves have not changed much. This relationship is particularly relevant during severe grid disturbances because the outer controller determines current references in the PLL reference frame when the relationship between the actual terminal voltage and the physical current determines the physical active and reactive powers. During normal operation, the inner current loop is generally much faster than the outer active- and reactive-power control loops, allowing I_d and I_q to approximately track their respective references.

The PLL also provides a useful connection between the dq -axis current components and the physical current phase. Under the usual space-vector convention, the current expressed in the stationary network reference frame can be represented as

$$\underline{I} = (I_d + jI_q) e^{j\theta_{\text{PLL}}} \quad (2.26)$$

Thus, the physical current angle is

$$\theta_I = \theta_{\text{PLL}} + \tan^{-1} \left(\frac{I_q}{I_d} \right) \quad (2.27)$$

with the exact sign and fixed phase offsets depending on the phase-current conventions adopted in the particular inverter model. The angle between the current and terminal voltage is

$$\phi_{IV} = \theta_I - \theta_V = \tan^{-1} \left(\frac{I_q}{I_d} \right) - \Delta\theta \quad (2.28)$$

Equation (2.28) shows that the current phase observed at the POC contains two distinct contributions: dq -current orientation and PLL synchronization state. A model that predicts only the current magnitude but neglects the PLL angle may reproduce the converter current limit while still producing an incorrect current phase and incorrect active and reactive power.

2.2.4 Balanced and Unbalanced Faults: Sequence Components

The short-circuit response of an IBR depends on whether the disturbance is balanced or unbalanced. The proposed framework focuses on balanced three-phase faults, for which the three-phase voltages and currents remain symmetrical and can be represented almost entirely using the positive-sequence component. This simplification is useful for analytical modeling because the fault response can be

studied using a single voltage and current phasor without explicitly modeling negative- or zero-sequence quantities.

For a balanced three-phase system, the phase currents and the phase voltages have equal magnitudes and are displaced by 120° :

$$\underline{I}_a = I \angle \theta_I \quad \underline{I}_b = I \angle (\theta_I - 120^\circ) \quad \underline{I}_c = I \angle (\theta_I + 120^\circ) \quad (2.29)$$

$$\underline{V}_a = V \angle \theta_V \quad \underline{V}_b = V \angle (\theta_V - 120^\circ) \quad \underline{V}_c = V \angle (\theta_V + 120^\circ) \quad (2.30)$$

Because the three phases remain symmetrical, their symmetrical-component decomposition contains only the positive-sequence component. In general, the phase currents can be transformed into zero-, positive-, and negative-sequence components according to

$$\begin{bmatrix} \underline{I}_0 \\ \underline{I}_1 \\ \underline{I}_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \underline{I}_a \\ \underline{I}_b \\ \underline{I}_c \end{bmatrix}, \quad a = e^{j2\pi/3} \quad (2.31)$$

For a balanced positive-sequence system,

$$\underline{I}_0 = 0, \quad \underline{I}_2 = 0 \quad (2.32)$$

and

$$\underline{I}_a = \underline{I}_1, \quad \underline{I}_b = a^2 \underline{I}_1, \quad \underline{I}_c = a \underline{I}_1 \quad (2.33)$$

The same relationships apply to the terminal voltage,

$$\underline{V}_0 = 0, \quad \underline{V}_1 = \underline{V} = V \angle \theta_V \quad \underline{V}_2 = 0, \quad (2.34)$$

Therefore, a balanced three-phase fault can be analyzed using only the positive-sequence voltage and current. For a GFL inverter, the positive-sequence terminal voltage is used by the PLL to define the local synchronous reference frame. As discussed in Section 2.2.3, if the PLL remains aligned with the positive-sequence voltage, $\theta_{\text{PLL}} \approx \theta_{V_1}$ then $V_d \approx V_1, V_q \approx 0$. These relationships connect the sequence-domain representation directly to the dq -axis model developed in the preceding subsections. For a balanced fault, I_d and I_q are simply the components of the positive-sequence fault-current vector in the PLL-oriented reference frame. The converter-current constraint can then be written directly as

$$I_1 = \sqrt{I_d^2 + I_q^2} \leq I_{\max}. \quad (2.35)$$

This makes the balanced fault simpler than an unbalanced fault. Because all three physical phase currents have equal magnitude, limiting the positive-sequence current to $I_{\max}$ simultaneously ensures that all three phase currents remain within the semiconductor current capability. This differs from an unbalanced fault, where positive- and negative-sequence currents can combine differently in each phase such that one phase may reach its current limit before the others.

Although a balanced fault does not introduce negative sequence, it does not necessarily follow that voltage phase can be ignored. A balanced three-phase fault can produce both a reduction in positive-sequence voltage magnitude and a positive-sequence voltage phase shift. The positive-

sequence terminal voltage and current can be expressed generally as

$$\underline{V}_1 = V_1 e^{j\theta_{V_1}} \quad \underline{I}_1 = I_1 e^{j\theta_{I_1}} \quad (2.36)$$

The positive-sequence complex power becomes

$$\underline{S}_1 = P + jQ = \underline{V}_1 \underline{I}_1^* \quad (2.37)$$

with

$$P = V_1 I_1 \cos(\theta_{V_1} - \theta_{I_1}) \quad Q = V_1 I_1 \sin(\theta_{V_1} - \theta_{I_1}) \quad (2.38)$$

Equation (2.38) shows that even during a completely balanced fault, the active and reactive powers depend on the relative voltage-current phase angle in addition to their magnitudes. The PLL and dq -current controller determine the orientation of the positive-sequence current relative to the faulted terminal voltage. Accordingly, a reduced-order model intended to reproduce both the magnitude and phase of the IBR fault current may need to account for $V_q, \theta_{V1}, P_{ref}, Q_{ref}, I_{max}, \theta_{PLL}$. This provides an important connection between the symmetrical-component representation and the analytical variables considered later in this thesis.

For completeness, asymmetrical faults such as line-to-line, single-line-to-ground, and double-line-to-ground faults introduce negative-sequence and, depending on the grounding path, potentially zero-sequence quantities. Under these conditions, $I_2 \neq 0$ and possibly $I_0 \neq 0$. When negative sequence is present, the phase currents become the vector combination of the positive- and negative-sequence currents, $\underline{I}_a = \underline{I}_1 + \underline{I}_2$ with corresponding 120° rotations in phases b and c . Unlike the balanced case, $\underline{I}_a, \underline{I}_b, \underline{I}_c$ need not be equal.

2.2.5 Grid-Following Fault-Current Limitation and Fault Recovery

As mentioned in Section 2.1.3, GFL inverters can be modeled as current-regulated resources whose outer control loops generate d - and q -axis current references according to active- and reactive-power objectives. As discussed in Section 2.2.3, the PLL establishes the rotating reference frame, while the inner current controller attempts to track these current references. Singh et al. [2] identify the PLL and current-control structure as fundamental components of generic GFL inverter models intended to reproduce system-level behavior. During normal operation, the requested currents generally remain below the converter capability, allowing the inverter to approximately maintain its reference active and reactive power. During a severe voltage disturbance, the current references may exceed the thermal capability of the semiconductor devices, requiring an explicit current-limiting mechanism.

Following Equations (2.35) and (2.20), the inverter cannot simultaneously maintain its original P_{ref} and Q_{ref} . Instead, the current must be modified according to the implemented current-limiting strategy. Entering current saturation represents a transition from a power-reference-dominated operating condition to a current-constrained operating condition. One possible current-limiting strategy is proportional vector scaling. The unconstrained current vector can be defined as

$$\underline{I}_{dq} = I_d + jI_q \quad (2.39)$$

A scaling coefficient can then be written as

$$k_I = \min \left(1, \frac{I_{\max}}{\sqrt{(I_d)^2 + (I_q)^2}} \right) \quad (2.40)$$

giving

$$I_{d,\lim} = k_I I_d \quad I_{q,\lim} = k_I I_q \quad (2.41)$$

This method retains the direction of the original dq -current vector while reducing its magnitude sufficiently to satisfy the converter current constraint. Practical inverter controllers may instead employ a priority-based limiter. For example, if reactive current is prioritized during low-voltage operation, the q -axis reference can first be constrained as

$$I_{q,\lim} = \text{sat}(I_q, -I_{\max}, I_{\max}), \quad (2.42)$$

with

$$\text{sat}(x, y, z) = \begin{cases} x & , y < x < z \\ y & , x \leq y \\ z & , x \geq z \end{cases} \quad (2.43)$$

The remaining current capability available to the active-current component is then

$$I_{d,\max} = \sqrt{I_{\max}^2 - (I_{q,\lim})^2} \quad (2.44)$$

and the d -axis current becomes

$$I_{d,\lim} = \text{sat}(I_d, -I_{d,\max}, I_{d,\max}) \quad (2.45)$$

The current constraint in Equation (2.35) does not by itself uniquely determine the inverter fault response. The control priority determines the location of the operating point on the current-limit circle and the active and reactive power delivered during the fault. Mahamedi et al. [16] demonstrate this broader principle using a fault-ride-through controller that combines synchronous-reference-frame current regulators with a phase-current limiter. Their limiter acts in the natural reference frame and limits the individual phase currents to a predefined threshold from the beginning of both balanced and unbalanced disturbances.

For the balanced three-phase faults considered in this thesis, $I_0 = I_2 = 0$, and all three phase currents have equal magnitude. The current-limit condition can be expressed directly in terms of the positive-sequence or dq -current magnitude, simplifying the current limiting problem:

$$|\underline{I}_a| = |\underline{I}_b| = |\underline{I}_c| = \sqrt{I_d^2 + I_q^2} \leq I_{\max} \quad (2.46)$$

Following Section 2.2.1, the GFL fault-current response can be represented as

$$\overbrace{P \approx P_{\text{ref}}, \quad Q \approx Q_{\text{ref}}}^{\text{prefault}} \longrightarrow \overbrace{\text{filter/control transient}}^{\text{fault inception}} \longrightarrow \overbrace{I \leq I_{\max}}^{\text{current-limited response}} \quad (2.47)$$

The current measured immediately following fault inception may differ from the value observed after the current-control and limiting mechanisms have settled. This distinction is important for the choice of sampling interval used to characterize IBR short-circuit current. When the fault is cleared, and terminal voltage begins to recover, the current required to satisfy the original power references decreases. In the simplest interpretation, the inverter leaves current saturation once it satisfies Equation (2.46). At this point, the limiter is not constraining the references and the controller may return toward $P \rightarrow P_{ref}$ and $Q \rightarrow Q_{ref}$. The finite dynamics of the PLL and current loop influence how quickly the inverter returns to the normal operating condition.

With I_{req} denoting the current magnitude required to satisfy active- and reactive-power references, the recovery process can be represented as

$$\underbrace{I = I_{max}}_{\text{faulted}} \longrightarrow \underbrace{I_{req} \downarrow}_{\text{voltage recovery}} \longrightarrow \underbrace{I_{req} < I_{max}}_{\text{limiter release}} \longrightarrow \underbrace{P, Q \rightarrow P_{ref}, Q_{ref}}_{\text{PLL/current-loop recovery}} \quad (2.48)$$

This distinction between current-limit release and complete controller recovery is important. Satisfying Equation (2.46) only indicates that the instantaneous current reference is again inside the allowable current circle; it does not imply that all internal controller states have already returned to their prefault conditions.

In practical inverter controllers, anti-windup, integrator clamping, state freezing, or related supervisory functions may also be used to prevent control integrators from accumulating excessive error while the converter is saturated. The specific controller studied in this thesis includes integrator-freeze behavior as one of the control features examined in Section 4.3.

2.2.6 Protection Implications and Short-Circuit Modeling Requirements

One of the most important differences between conventional and inverter-dominated fault behavior is the limited over-current capability of power-electronic converters. As established in Sections 2.2.2 and 2.2.5, the inverter current must satisfy Equation (2.46). Once the converter enters current-limited operation, $I \approx I_{max}$, and further reduction in terminal voltage does not necessarily result in the proportional increase in fault current that would be expected from a fixed source impedance. The fault severity and measured current magnitude are not necessarily related.

The time at which fault current is evaluated also becomes important. As shown by Mahmud et al. [5] and Equation (2.47), a short-circuit model intended for protection applications should identify which portion of the fault waveform is being represented. Cheng et al. [6] directly examine this issue by comparing generic IBR models used in conventional short-circuit programs with detailed manufacturer EMT models. Their work is motivated by the expectation that increasing IBR penetration will alter bulk-system short-circuit characteristics and may require changes to protection studies and relay settings. This creates a practical modeling trade-off. A detailed EMT representation can include converter switching or averaged converter dynamics, filters, PLL behavior, current controllers, outer control loops, limiters, and supervisory functions. Such models provide a high level of detail but are considerably more complex than the source models traditionally used in large short-circuit studies. On the other hand, an excessively simple constant-current or impedance-based representation may omit controller-dependent relationships that influence the current seen by protection equipment.

Mahmud et al. [5] demonstrate that an intermediate approach is possible. Their reduced-order parameterized short-circuit model separates the inverter response into regions analogous to the initial, transient, and steady-state portions of the fault waveform and identifies the components that dominate each region. This supports the central motivation for using a reduced analytical model instead of attempting either to reproduce every internal converter state or to assume a fixed fault-current source. Based on the literature reviewed in Section 2.2, a useful reduced-order short-circuit model for the balanced GFL conditions considered in this thesis should satisfy several requirements.

First, it should reproduce the effect of terminal-voltage magnitude. Before saturation, maintaining approximately constant power produces an inverse relationship between current demand and voltage as in Equation (2.20) whereas after saturation, $I \leq I_{\max}$. Second, the model should retain the prefault operating point. Because P_{ref} and Q_{ref} determine the original d - and q -axis current requirements, two inverters experiencing the same fault voltage need not produce identical fault currents if their prefault operating conditions differ. Third, it should reproduce the current limit and current-priority behavior, Equation (2.35), because saturation causes the actual P and Q to depart from their original references. Fourth, the model should account for voltage and current phase. As demonstrated in Sections 2.2.3 and 2.2.5, PLL state and dq -current allocation can alter the physical current angle. This phase is necessary to reproduce P , Q , and protection quantities such as apparent impedance.

2.2.7 Utility Protection Requirements — Hydro One

From a transmission-utility perspective, the purpose of a short-circuit model is not simply to calculate the maximum fault-current magnitude, but to provide electrical quantities with sufficient accuracy to evaluate whether protection systems detect, discriminate, and clear faults correctly. Hydro One’s current Transmission Generation Interconnection Requirements, dated March 6, 2025, explicitly identify short-circuit and protection-impact studies as part of the interconnection assessment and require detailed protective-relay coordination studies for new generation facilities [17]. For IBR plants, Hydro One further requires submission of a detailed EMT model that accurately represents the plant controls and protection functions for compliance and performance studies. A useful protection-oriented reduced model must provide I_{fault} and the voltage and current phasors required by the relevant relay functions. In particular, the apparent impedance measured by a distance element can be expressed as

$$\underline{Z}_{app}(t) = \frac{\underline{V}(t)}{\underline{I}(t)} = \frac{|V(t)|}{|I(t)|} e^{j[\theta_V(t) - \theta_I(t)]} \quad (2.49)$$

Equation (2.49) demonstrates why prediction of fault-current magnitude alone is insufficient: an error in θ_I directly changes both the magnitude and angle of the impedance presented to the relay, even when $|I|$ is predicted correctly.

This requirement is particularly evident in Hydro One’s Protection Planning Standard for Transmission Generation Connections [18]. Although the examples in that standard use SGs, Hydro One explicitly states that the connection requirements apply equally to synchronous and non-synchronous generation. The standard requires generation-facility line protection to detect phase and ground faults on the associated HV transmission line and discusses how fault current modifies the apparent impedance measured by distance protection. For a remote-end fault, the strong Hydro One system

may contribute more fault current than the connected generation facility, causing the impedance observed from the generation terminal to become comparatively large; protection reach and coordination must be evaluated for the actual source and network configuration. This becomes even more significant for inverter-based generation. Hydro One specifically notes that variable-capacity wind or solar generation may provide insufficient current input for the facility protection to detect a fault and that, for a weak source, voltage-based line protection may be required when the apparent impedance becomes too large. Thus, the traditional assumption that a more severe fault necessarily produces a sufficiently large current for dependable protection cannot be applied directly to an IBR.

The quantities required from an IBR short-circuit model also depend on the protection function and on the time interval being studied. An over-current element is principally influenced by whether $|I(t)| \geq I_{\text{pickup}}$ whereas distance protection and other phase-sensitive quantities require the relative voltage-current phase relationship. In addition, the relevant quantities are time dependent, as illustrated by Equation (2.47). Hydro One’s protection requirements similarly emphasize coordination, high-speed fault clearing, transfer-trip and blocking schemes, meaning that the value returned by a short-circuit calculation must correspond to a clearly defined portion of the disturbance rather than an unspecified “fault current” [18]. Hydro One also requires disturbance monitoring, including fault recording and, where required, PMUs, so that the sequence and timing of system events can subsequently be reconstructed.

2.3 Existing Models

2.3.1 Constant-Current and Constant-Power Models

One of the simplest representations of an IBR during a fault is an ideal controlled current source. Under this approximation, the inverter injects a prescribed current independently of the terminal voltage,

$$\underline{I}_{\text{IBR}} = I_{SC} e^{j\theta_I} \quad (2.50)$$

where I_{SC} and θ_I are the assumed fault-current magnitude and phase, respectively. In some short-circuit studies, I_{SC} is selected directly from the inverter over-current capability, such that $I_{SC} \approx I_{max}$. Choi et al. [19], for example, represented an IBR as a constant-current source supplying approximately 1.2 pu fault current for steady-state short-circuit calculations. This representation is computationally attractive and can approximate the sustained response of an inverter after its current limiter has become dominant. However, a constant-current representation satisfies only part of the modeling requirements established in Section 2.2.6. Because Equation (2.50) contains no dependence on terminal voltage or prefault P/Q , it cannot reproduce the voltage-dependent transition into current-limited operation. Also, assigning the same I_{SC} for every operating condition removes the influence of P_{ref} and Q_{ref} .

A constant-power representation retains more of the converter’s normal GFL control behavior. From the complex-power relationship at the point of connection,

$$\underline{S}_{\text{ref}} = P_{\text{ref}} + jQ_{\text{ref}} = \underline{V} \underline{I}^* \quad (2.51)$$

the current required to maintain the referenced active and reactive powers is

$$\underline{I}_{CP} = \left(\frac{P_{ref} + jQ_{ref}}{\underline{V}} \right)^*, \quad I_{CP} = \frac{\sqrt{P_{ref}^2 + Q_{ref}^2}}{|\underline{V}|} \quad (2.52)$$

Equivalently, when the dq reference frame is aligned with the terminal voltage, the corresponding current references are approximately $I_d = P_{ref}/V$ and $I_q = -Q_{ref}/V$. These relationships are consistent with the controlled-current interpretation of a GFL inverter in Section 2.1.3 and with the voltage-dependent power-transfer relationships reported by Ravanji et al. [3]. Unlike the constant-current approximation, Equation (2.52) directly retains both the prefault operating point and the terminal-voltage magnitude: two inverters exposed to the same fault voltage may request different currents if their prefault P_{ref} and Q_{ref} differ. It also explains the approximately inverse voltage-current characteristic expected while the outer power controller is still attempting to preserve its reference power. The principal limitation of the constant-power model becomes apparent as the terminal voltage decreases. Equation (2.52) predicts $I_{CP} \rightarrow \infty$ as $V \rightarrow 0$, whereas a practical inverter must satisfy the current constraint Equation (2.35).

Once the unconstrained current request exceeds I_{max} , the inverter cannot simultaneously maintain $P = P_{ref}$ and $Q = Q_{ref}$, and its current limit; its actual response instead depends on the implemented active- or reactive-current priority. A more realistic algebraic representation must be at least piecewise,

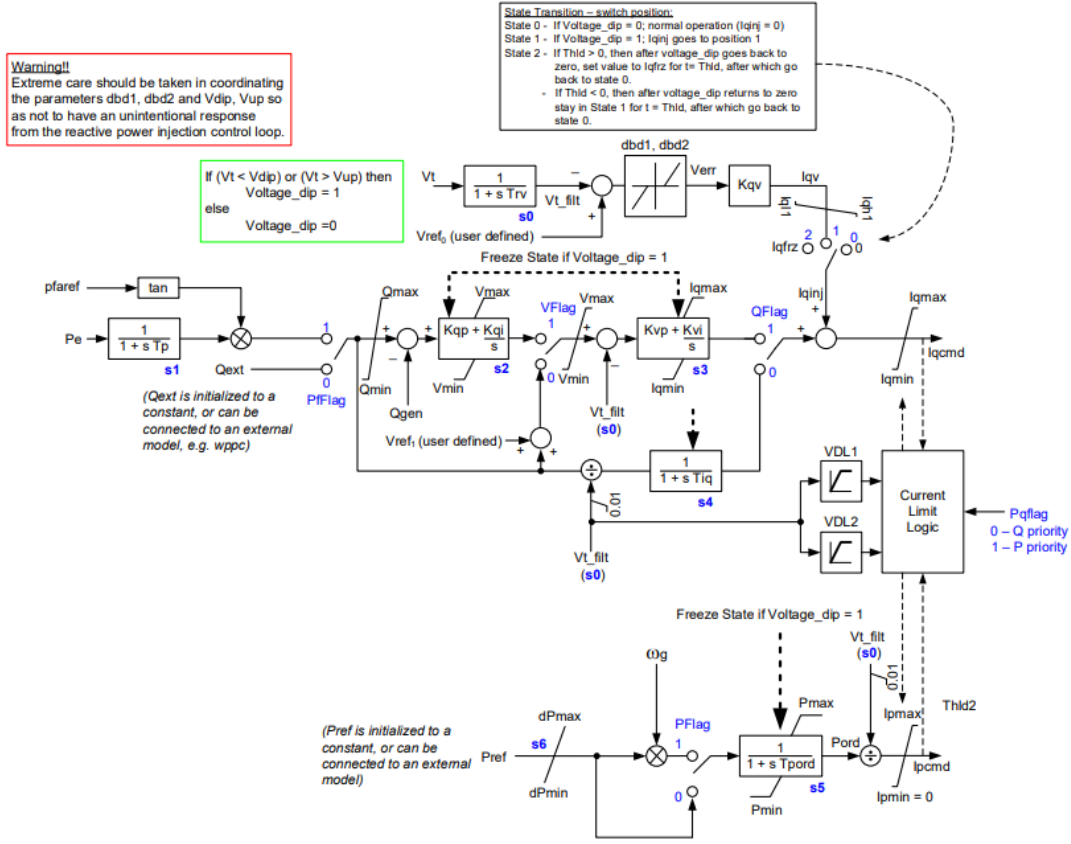
$$\underline{I} = \begin{cases} \underline{I}_{CP}, & |\underline{I}_{CP}| \leq I_{max}, \\ \mathcal{L}(\underline{I}_{CP}, I_{max}, \text{current priority}), & |\underline{I}_{CP}| > I_{max}. \end{cases} \quad (2.53)$$

where $\mathcal{L}(\circ)$ represents the converter-specific limiting and current-allocation logic. The voltage at which the constant-power request first reaches the current limit is

$$V_{sat} = \frac{\sqrt{P_{ref}^2 + Q_{ref}^2}}{I_{max}} \quad (2.54)$$

Equation (2.53) gives a useful physical division between an unsaturated region, in which approximately constant-power behavior may be possible, and a saturated region in which the converter must depart from its original P - and Q -references. However, the current-limited constant-power formulation remains incomplete for protection-oriented short-circuit calculations because it does not necessarily reproduce the physical current phase. Using the complex voltage in Equation (2.52) implicitly assumes that the current reference responds directly to the terminal-voltage phasor, whereas an actual GFL inverter generates the current vector in a PLL-oriented reference frame.

Constant-current and constant-power models describe two useful limiting behaviors: the former approximates a sustained saturated current source, whereas the latter approximates the unsaturated power-regulation region. Neither independently satisfies all four requirements identified in Section 2.2.6, motivating a reduced-order model that preserves voltage dependence, prefault operating condition, current-limiting and priority behavior, and current phase.

Figure 2.4: EPRI $REEC_A$ Model of an IBR as Highlighted in [4].

2.3.2 WECC and EPRI Generic Positive-Sequence Models

The WECC/EPRI generic renewable-energy models provide a more detailed positive-sequence representation of IBRs than the constant-current and constant-power approximations discussed previously. The models are modular: the electrical-control model (REEC) (Fig. 2.4) converts plant-level active- and reactive-power control commands into active- and reactive-current references, and the generator/converter model (REGC) (Fig. 2.5) provides the electrical interface between the converter and the network [4, 20]. EPRI describes the corresponding signal flow as

$$(P_{\text{ref}}, Q_{\text{ref}}, V_t) \xrightarrow{\text{REEC}} (I_{p,\text{cmd}}, I_{q,\text{cmd}}) \xrightarrow{\text{REGC}} I_{\text{IBR}}. \quad (2.55)$$

The electrical-control layer preserves the prefault operating condition rather than assigning every inverter the same fault-current contribution. In the EPRI implementation, P_{ref} and the reactive-power reference Q_{ext} are initialized as model references and can subsequently be controlled by the plant controller [4].

The REEC models also introduce explicit voltage dependence and converter-current constraints. Under normal operation, the active- and reactive-current commands retain behavior analogous to the constant-power relationships developed in Section 2.2.2,

while adding generic proportional-integral inner-current controllers and an explicit PLL, including the PLL gains $K_{p,PLL}$, $K_{i,PLL}$, frequency limits, and a low-voltage threshold V_{dip} below which the PLL state can be frozen [4]. The synchronization dynamics can be represented as

$$\Delta\theta = \theta_V - \theta_{PLL}, \quad \dot{\theta}_{PLL} = \omega_0 + \Delta\omega_{PLL} \quad (2.57)$$

so that the orientation of the commanded dq -current vector is not assumed to follow the terminal-voltage angle instantaneously. This feature is particularly significant for the present work because Section 2.2 showed that the physical current phase depends on both I_d/I_q allocation and PLL state, and affects P , Q , and protection quantities even when current magnitude is reproduced correctly.

The WECC/EPRI models satisfy more of the short-circuit modeling requirements identified in Section 2.2.6 than either constant-current or constant-power representations. They retain voltage-magnitude dependence, the initialized P/Q operating point, total current capability, and active/reactive current priority. Nevertheless, they remain generic positive-sequence phasor-domain dynamic models instead of detailed protection-oriented converter models. Under weak-grid conditions, *REGC_C* reveals PLL and current-loop interactions that the simpler models cannot reproduce and may indicate when a detailed EMT investigation is necessary. Thus, *REGC_C* closes the gap between conventional positive-sequence and EMT representations as its response depends on generic parameterization and does not necessarily reproduce manufacturer-specific nonlinear control, limiting, or state-freezing behavior.

2.3.3 EMT Models and the Accuracy–Complexity Tradeoff

At the highest-fidelity end of the IBR modeling spectrum are EMT models, which represent the three-phase electrical network directly in the time domain rather than assuming a fundamental-frequency positive-sequence phasor. PSCAD, and its EMTDC simulation engine, calculates instantaneous electromagnetic responses, allowing the converter, transformer, filters, network, and controls to be represented at finer temporal resolution than conventional short-circuit or positive-sequence programs. A detailed IBR EMT model can explicitly retain the PLL, dq -current controllers, outer P/Q control loops, converter current limiter, active/reactive current priority, sequence extraction, protection functions, and supervisory logic such as integrator freezing. Its behavior may be written generally as a nonlinear time-domain system,

$$\dot{\underline{x}} = f(\underline{x}, \underline{V}_{abc}, \underline{u}, t), \quad \underline{I}_{abc} = g(\underline{x}, \underline{V}_{abc}, \underline{u}, t) \quad (2.58)$$

where $\underline{x}$ contains the internal electrical and controller states and $\underline{u}$ contains operating references and control settings. Unlike an algebraic short-circuit model of the form $I = f(V, P_{ref}, Q_{ref})$, Equation (2.58) allows the current at a particular instant to depend on the history and internal state of the inverter. This distinction attempts to model the inverter behavior during faults where the initial current transient, limiter activation, PLL motion, and subsequent recovery do not need to occur simultaneously.

This additional state information enables EMT models to satisfy the four protection-oriented modeling requirements identified in Section 2.2.6 more naturally than the reduced representations. The prefault power-flow solution establishes the initial terminal-voltage magnitude and phase and the corresponding P_{ref} , Q_{ref} , I_d , I_q , and controller states before the disturbance; PSCAD itself provides

initialization mechanisms intended to establish the appropriate prefault bus voltages, angles, and active/reactive power flows before a fault is applied. Once the fault occurs, the instantaneous solution can reproduce voltage-magnitude dependence, current saturation and d/q -priority transitions, and the evolution of the physical current angle. Wu et al. [21] show that the PLL and inner current loop can become dynamically coupled during severe disturbances, reinforcing the importance of retaining these states when their effect on the current phasor is significant. Similarly, Cheng et al. [6] use detailed manufacturer PSCAD EMT models as the reference against which generic short-circuit representations in ASPEN, CAPE, and CYME are evaluated, specifically examining whether the simplified models reproduce manufacturer-dependent behavior such as sequence-current injection.

The increased fidelity of EMT simulation is obtained at a considerable modeling and computational cost. EMT simulation solves the instantaneous three-phase network with small integration time steps and can require detailed representations of converter controls, plant controllers, transformers, cables, protection functions, and network equivalents. NERC distinguishes detailed vendor-specific EMT models and reduced vendor-specific EMT models from generic positive-sequence models, noting that the former preserve more equipment and control detail but are commonly proprietary and may require OEM participation or nondisclosure arrangements [8]. Furthermore, the computational burden increases rapidly as the study network and number of IBRs increase. EPRI illustrates this tradeoff directly: under relatively strong-grid conditions, $REGC_A$, $REGC_B$, and $REGC_C$ positive-sequence models agree reasonably well with an EMT representation, whereas under weaker conditions the EMT simulation reveals oscillatory behavior associated with fast converter dynamics that the simpler generic representations reproduce with varying success [4]. NERC has similarly documented increasing discrepancies between positive-sequence and EMT simulations as system strength decreases, including differences in voltage and reactive-power response following faults [8].

For the above reasons, the PSCAD EMT model used in this thesis (Section 3.1) is adopted as the benchmark representation from which the reduced empirical short-circuit model is derived and validated. Importantly, EMT simulation should not be interpreted as intrinsically exact; NERC explicitly cautions that EMT accuracy depends on model development, assumptions, parameterization, and verification, and recommends that models are sufficiently validated before being relied upon for system studies [8]. In the present work, the objective is not to replace EMT for detailed equipment studies, but to reproduce the protection-relevant outputs of the EMT response with sufficient accuracy for rapid short-circuit calculations.

2.3.4 Comparison of Existing Models

The models reviewed in Section 2.3 form a hierarchy between computational simplicity and representation of converter control. The comparison in Table 2.2 is made using the four requirements derived in Section 2.2.6: (1) terminal-voltage dependence, (2) retention of the prefault P/Q operating point, (3) current-limit and current-priority behavior, and (4) current phase and synchronization effects. No conventional simplified model satisfies all four requirements with the same fidelity as a detailed EMT model. The objective of the empirical framework developed in the following chapters is to retain the protection-relevant nonlinear relationships observed in PSCAD while avoiding the model complexity and computational burden of complete EMT simulation.

Table 2.2: Comparison of Existing IBR Short-Circuit Modeling Approaches

Modeling approach	Terminal-voltage dependence	Prefault $P_{\text{ref}}, Q_{\text{ref}}$	Current limit / priority	Current phase / PLL	Principal limitation
Constant-current model	×	×	Partial — can impose I_{max} , but not allocation	×	Represents mainly the saturated region
Constant-power model	✓	✓	×	Partial — algebraic voltage angle, no independent PLL state	$I \rightarrow \infty$ as $V \rightarrow 0$
Current-limited constant-power model	✓	✓	✓ if priority is specified	Partial	Synchronization/control dynamics remain simplified
WECC/EPRI REGC/REEC	✓	✓	✓	Partial for earlier models; improved in REGC_C	Generic positive-sequence representation
Detailed PSCAD EMT model	✓	✓	✓	✓	High computational and model-information requirements
Proposed empirical framework	✓ target	✓ target	✓ target	✓ target	Reduced representation must be validated against EMT

2.4 Research Gaps

2.4.1 Neglect of Voltage Phase

Many simplified IBR short-circuit models characterize a fault primarily through the terminal-voltage magnitude while neglecting its phase after the fault or assuming that the PLL remains instantaneously synchronized with the terminal-voltage phasor. Although this assumption is reasonable during normal synchronized operation such that the prefault voltage phase is synchronized to the grid, a fault can produce both a voltage-magnitude depression and a phase shift while the PLL responds with finite dynamics. As established in Section 2.2.3, two fault conditions with similar fault voltage magnitudes may not necessarily produce the same current phasor if their fault voltage angles and PLL states differ. This limitation is particularly relevant to protection-oriented modeling because quantities such as current angle and apparent impedance depend on the relative phase between voltage and current. Lyu et al. [15] emphasize the importance of current-angle behavior for protection. Even for the balanced three-phase faults considered in this thesis, the absence of negative sequence does not eliminate positive-sequence voltage-phase shift.

2.4.2 Simplification of PLL Dynamics and Low-Voltage Controller State

A second limitation concerns the treatment of the PLL during severe voltage depressions. Most algebraic short-circuit models contain no PLL state at all, while simpler positive-sequence models effectively assume that synchronization remains sufficiently accurate throughout the disturbance. More detailed generic and EMT models can represent PLL dynamics, but the resulting behavior depends on the particular implementation and low-voltage supervisory logic. A PLL cannot obtain the same reliable phase information when the measured terminal voltage collapses. Practical converter controls may clamp integrators, freeze selected controller states, inhibit portions of the PLL, or use related low-voltage logic.

This behavior is represented by introducing a voltage-dependent supervisory state,

$$\gamma(V) = \begin{cases} 1, & V \geq V_{\text{freeze}}, \\ 0, & V < V_{\text{freeze}}. \end{cases} \quad \dot{\underline{x}}_{\text{PLL}} = \gamma(V) f_{\text{PLL}}(\underline{x}_{\text{PLL}}, \underline{V}) \quad (2.59)$$

where $\underline{x}_{\text{PLL}}$ denotes the PLL states. Equation (2.59) is intentionally generic because different implementations may freeze only an integrator, clamp the estimated frequency, or retain an angle derived from the last reliable synchronization state. In each case, the important consequence is that θ_{PLL} acquires memory of the operating condition at which the freeze occurred. The fault current can depend on the instantaneous value of V and also on the controller state immediately before entering the low-voltage region [21]. A reduced short-circuit formulation based only on instantaneous terminal voltage cannot naturally reproduce this history-dependent response, motivating explicit investigation of V_{freeze} , frozen controller states, and their influence on current magnitude and phase.

2.4.3 Simplified or Missing Current-Limiting Behavior

A third modeling gap is the treatment of converter current limitation. This limitation should again be stated carefully: modern WECC/EPRI models, sequence-domain controllers, and EMT models

can explicitly include current limiting, but many elementary short-circuit formulations either omit it entirely or approximate the inverter as a fixed current source once saturation occurs. Neither approach captures the complete transition. A constant-power model in Equation (2.20) becomes unphysical as $V \rightarrow 0$, whereas simply imposing $I = I_{max}$ gives no information regarding the allocation of this finite capability between the active- and reactive-current components. Equation (2.35) alone does not uniquely determine the resulting fault current because the control priority determines where the inverter operates on the current-limit circle (Equation (2.42)).

2.4.4 Inability to Capture Nonlinear and Piecewise Fault Behavior

IBR fault behavior is inherently nonlinear and can change operating regime during a single disturbance. Conventional linearized or single-equation representations are attractive because of their simplicity, but they generally describe only a restricted operating region. For example, before the converter reaches its current limit, approximately constant-power control produces $I \propto 1/V$; after saturation, the magnitude may approach I_{max} ; and once the voltage crosses a low-voltage control threshold, PLL or controller-state freezing can introduce an additional change in the relationship between voltage and current:

$$\underline{I}_{IBR} = \begin{cases} f_{\text{normal}}(\underline{V}, P_{\text{ref}}, Q_{\text{ref}}), & V \geq V_{\text{freeze}}, \text{ not current limited,} \\ f_{\text{freeze}}(\underline{V}, \underline{x}_{\text{freeze}}, I_{\text{max}}), & V < V_{\text{freeze}}, \text{ not current limited,} \\ f_{\text{lim}}(\underline{V}, P_{\text{ref}}, Q_{\text{ref}}, I_{\text{max}}, \text{ priority}), & \text{ current limited.} \end{cases} \quad (2.60)$$

where $\underline{x}_{\text{freeze}}$ represents the retained controller state at the low-voltage transition.

The model developed explicitly distinguishes operation above and below V_{freeze} , investigates voltage phase near both low voltage and the freezing boundary, and examines current limitation and frozen-controller behavior as separate physical mechanisms. The central research gap is summarized as the absence of a computationally simple analytical model that preserves these nonlinear transitions while simultaneously retaining voltage magnitude and phase, prefault P/Q , current limitation and priority, and low-voltage controller state. The proposed empirical framework addresses this gap by using PSCAD EMT results to identify the dominant operating regions of a GFL converter and then constructing a reduced piecewise formulation whose parameters remain physically interpretable.

Chapter 3

Test System and Simulation Setup

3.1 PSCAD Model

The simulation study in this thesis is based on the EPRI EMT inverter model implemented in PSCAD [1] as the high-fidelity benchmark from which a simpler empirical short-circuit representation is developed. The model provides a detailed representation of the inverter and its associated controls, allowing its effects to be represented during grid faults [1]. Although the EMT implementation provides the control detail needed to investigate nonlinear fault response, its small simulation time steps and detailed internal states require more computational effort than conventional phasor-domain calculations.

3.1.1 Top Level EPRI Inverter Model

Figure 3.1 shows the top-level configuration of the EPRI inverter model used throughout this thesis. The model represents a utility-scale photovoltaic plant connected to the grid through a 120/33 kV three-phase transformer and a detailed Solar Farm subsystem. The Solar Farm receives P_{ref} , Q_{ref} , and V_{ref} as its principal operating references and contains the converter and associated control systems described in the following sections. The system is configured on a 60 Hz, 33 kV, 100 MW base, providing the reference quantities used for the per-unit analysis in this work.

The EPRI implementation supports several GFL and GFM control configurations through the $Qflag$, $Vflag$, and controller or W -evaluation settings summarized in Table 3.1. With $Qflag = 0$, the model operates as a conventional GFL inverter, the focus of this thesis, using a synchronous-reference-frame phase-locked loop (SRF-PLL) and reactive-power injection control. Other configurations allow active- and reactive-power regulation with frequency support or GFM operation using PLL-based control, droop control, a virtual synchronous machine (VSM), or dispatchable virtual oscillator control (dVOC) [1]. Holding the operating mode fixed also allows the effects of terminal voltage, P_{ref} , Q_{ref} , current limitation, and controller-state freezing to be examined independently of changes in the fundamental inverter-control architecture.

Several control limits are also fixed to define the fault-response characteristics investigated in this thesis. Reactive current is given priority, and the maximum inverter-current magnitude is set to $I_{\text{max}} = 1.2$ pu. The model additionally contains low- and high-voltage state-freeze thresholds of 0.8 pu and 1.3 pu, respectively. When the relevant measured dq -frame voltage leaves this permitted

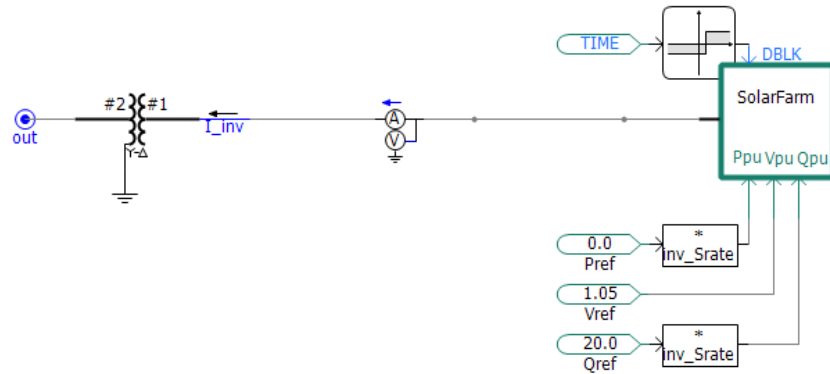


Figure 3.1: Block Diagram of the EPRI Inverter Model [1].

Table 3.1: Summary of GFL and GFM operating configurations in the EPRI inverter model [1].

Operating mode	$Qflag$	$Vflag$	Controller / W evaluation	Description
GFL	0	–	SRF-PLL	Standard reactive-power (Q) injection GFL configuration.
GFL	1	1	SRF-PLL	Controls active- and reactive-power injection, with some inverter-level frequency support.
GFM	1	0	SRF-PLL	PLL-based grid-forming configuration with voltage-forming behavior.
GFM	1	0	Droop	Non-PLL-based grid-forming operation using droop control.
GFM	1	0	VSM	Non-PLL-based grid-forming operation using a virtual synchronous machine (VSM) controller.
GFM	1	0	dVOC	Non-PLL-based grid-forming operation using dispatchable virtual oscillator control (dVOC).

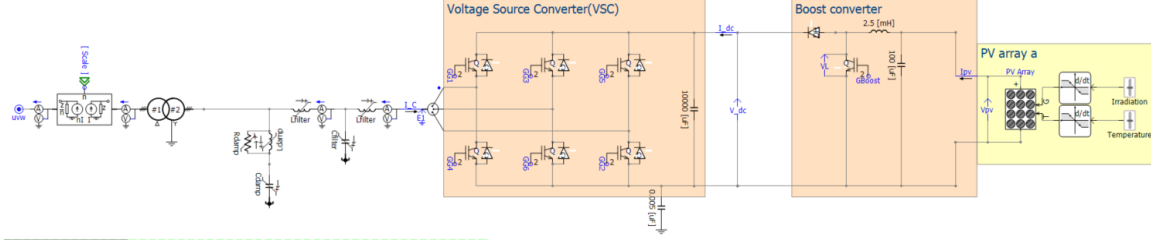


Figure 3.2: Block Diagram of the EPRI Inverter Model of the Solar Farm [1].

range, selected controller-integrator states are prevented from continuing their normal evolution, so the inverter does not regulate P and Q in the same manner as it does within the normal-voltage region. The inverter also has a parameter for over-current, which permits the current magnitude to go above the specified maximum current magnitude for 0.2 seconds during a fault. This setting is a requirement for IEEE 2800 standard [7] to prevent massive power loss on the grid right after a fault. The important characteristics of the EPRI inverter model are summarized in Table 3.2.

Table 3.2: EPRI Top Level Model Summary

Base Voltage	Base Power	$I_{\max}$	V_{freeze}
33 kV	100 MW	1.2 pu	0.8 pu

3.1.2 PV Solar Farm Inverter Level

The main inverter implementation is contained within the Solar Farm subsystem shown in Fig. 3.2. At the inverter level, the model can be divided into five principal sections: the photovoltaic (PV) array, boost converter, voltage-source converter (VSC), LCL filter, and scaling three-phase transformer. The PV array provides the DC-side energy source and is driven by temperature and solar-irradiance inputs, producing the corresponding DC voltage and current. In the EPRI implementation, this PV array represents one portion of a larger solar facility, with the output subsequently scaled to represent the aggregate response of multiple PV units within the plant.

The DC output of the PV array is supplied to the boost converter, whose controller regulates its duty cycle so that the converter output follows the specified DC-voltage reference. The regulated DC voltage and current are then supplied to the VSC, which forms the electrical interface between the DC source and the three-phase AC system. Using measurements and reference quantities generated by the inverter control system, the VSC determines the required phase-voltage references and produces the switching pulses for phases A, B, and C. Because these switching commands ultimately determine the AC voltage and current injected by the inverter, the VSC and its associated control system have a major influence on the inverter response during both normal operation and grid disturbances. The internal VSC controller is examined in greater detail in Section 3.2. Following the VSC, the generated three-phase waveform passes through the LCL filter before reaching the external network. Finally, the scaling three-phase transformer represents the aggregation and interconnection of multiple physical PV inverter units, allowing the detailed inverter-level response to be represented at the utility-scale plant rating used in the study.

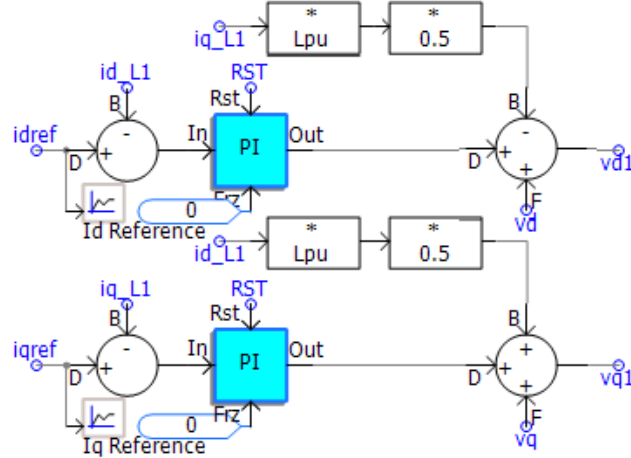


Figure 3.4: Block Diagram of the EPRI Inverter Model for the VSC Inner Current Controllers [1].

thereby aligning the d -axis with the terminal-voltage vector. The resulting phase and frequency are then used in subsequent dq transformations, closing the synchronization feedback loop. During a fault, changes in terminal-voltage magnitude or phase affect both the PLL reference frame and the current-reference controller. The voltage-freeze logic can prevent the integral states from continuing to evolve once the specified voltage boundary is crossed.

3.2.2 Inner Current Loop, Phase Voltage References, and Pulse Control

The $I_{d,\text{ref}}$ and $I_{q,\text{ref}}$ references generated by the outer dq -current reference controller are passed to the inner current-control loop shown in Fig. 3.4. The controller compares these references with the measured d - and q -axis currents flowing through the LCL filter and processes the resulting current errors using PI controllers. The controller also includes compensation terms associated with the d - and q -axis coupling across the filter, producing the referenced converter voltages V_d and V_q . Because this inner loop acts directly on current error, it is responsible for forcing the physical inverter current to follow the references established by the slower outer control system. Fig. 3.4 represents the link between the desired dq -axis current response and the converter voltage required to produce that current.

The referenced V_d and V_q are subsequently passed to the phase-voltage reference controller shown in Fig. 3.5. Using the phase reference supplied by the synchronization system, the controller performs the inverse dq -to- abc transformation to obtain the three phase-voltage references required by the VSC. The referenced voltage magnitude is also constrained between approximately 0.2 pu and 1.15 pu in the EPRI implementation, preventing the controller from requesting unrealistically low or high converter voltages. Fig. 3.5 converts the rotating-frame control quantities into the physical three-phase voltage commands used by the converter. These V_{abc} references are then used by the VSC pulse-firing logic to generate the switching signals for the semiconductor devices, completing the control path from P_{ref} and Q_{ref} to the actual three-phase voltage and current injected into the network.

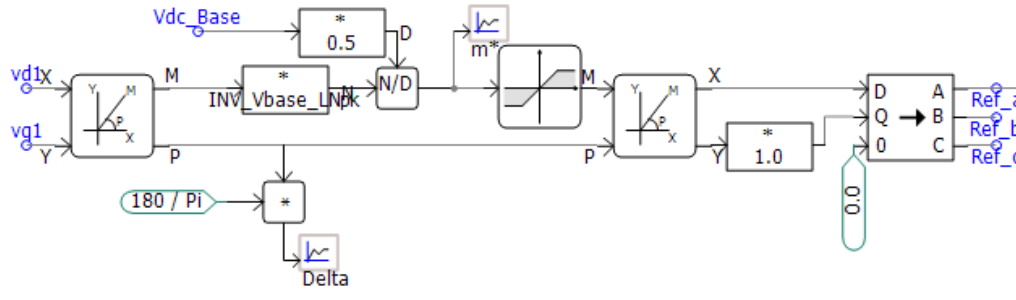


Figure 3.5: Block Diagram of the EPRI Inverter Model for the VSC Phase Voltage Reference Controller [1].

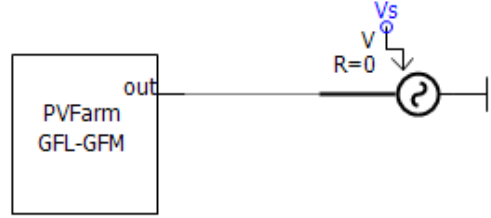


Figure 3.6: Simple Inverter Short-Circuit Testbench Setup. PVFarm contains the top-level EPRI EMT representation of Fig. 3.1.

3.3 Simulation Cases

The simulation cases are designed to isolate the variables that most strongly influence the short-circuit response of the GFL inverter and to generate the dataset required for development of the empirical model. The principal variables considered in this study are the terminal-voltage magnitude and phase, the prefault active- and reactive-power references P_{ref} and Q_{ref} , the prefault voltage, and the inverter control constraints described in Section 3.2. To study these effects independently of interactions with a larger network, the initial data collection is performed using the simple testbench shown in Fig. 3.6, in which the EPRI inverter is connected directly to a controllable voltage source. This arrangement is consistent with the controlled-current-source interpretation of a GFL inverter: the external source establishes the voltage magnitude, phase, and frequency seen by the inverter, while the inverter controller determines the current injected in response to those conditions.

For each simulation, the system is first allowed to reach a stable prefault operating condition before a disturbance is introduced. The source voltage is initially established near its nominal 120 kV value and then changed abruptly in magnitude and, where applicable, phase to reproduce the voltage disturbance that would be observed at the inverter terminal during a grid fault. Instead of modeling different physical fault impedances for every case, this approach directly sweeps the resulting terminal-voltage condition, allowing the inverter response to be characterized over a wide range of voltage depressions. The simulations also vary the prefault voltage, including approximately 0.9, 1.0, and 1.1 pu operating conditions, while P_{ref} and Q_{ref} are swept over the required operating range. The resulting terminal V , P , Q , current magnitude, and current phase are recorded after the initial fault transient so that the sustained controller response can be identified. These sweeps form the empirical dataset used in Sections 4 and 5 to determine which variables dominate the fault

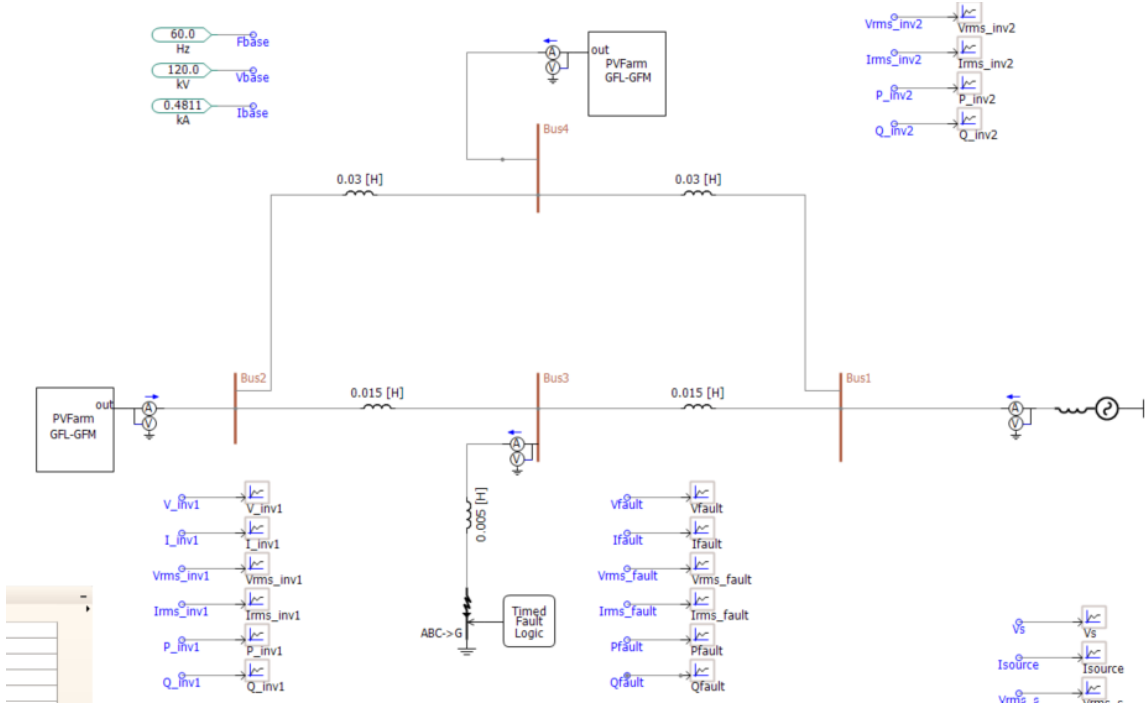


Figure 3.7: 3-Bus Network Testbench for Model Verification.

response and to identify the parameters of the proposed analytical model.

After the analytical relationships are obtained from the single-inverter testbench, the model is evaluated under a more realistic interconnected-network condition using the system shown in Fig. 3.7. This verification system contains two IBRs operating at different prefault conditions together with a SM and an interconnected transmission network. The same disturbance is solved both with the detailed PSCAD EMT representation and with an analytical power-flow implementation containing the proposed inverter model. Unlike the isolated testbench of Fig. 3.6, the inverter terminal voltages in this case are not prescribed directly; instead, they result from the interaction between the inverter current injections and the network. This provides a more stringent verification of the model because the calculated inverter current changes the network voltage, which in turn changes the current predicted by the empirical formulation until a self-consistent solution is obtained.

Finally, the study extends beyond the small verification network to examine the numerical scalability and convergence of the proposed formulation. A synthetic 4000-bus power-flow network containing a fault is used to determine whether the nonlinear voltage-dependent inverter model can be incorporated into a larger sparse network calculation without preventing convergence.

3.4 Data Collection

3.4.1 Prefault Data

To illustrate the data-collection procedure and verify that the inverter is correctly initialized before the disturbance, a representative simulation case is considered with $P_{\text{ref}} = 0.8$ pu, $Q_{\text{ref}} = 0.2$ pu, and no imposed change in voltage phase. The external source is initially maintained at 120 kV and

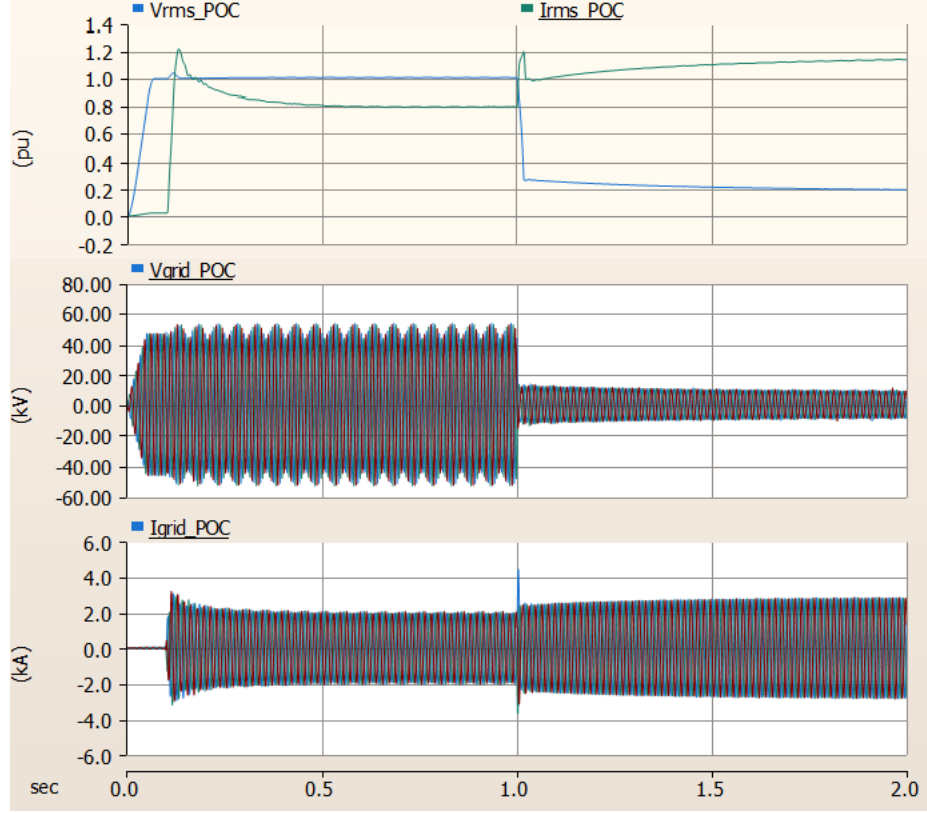


Figure 3.8: Time Domain Voltage and Current RMS Values with their Associated abc V and I Waveforms with Fault Introduced at $t=1$ s.

is reduced to 30 kV at $t = 1$ s to represent the fault-induced voltage depression. Because the inverter is connected through a 120/33 kV transformer and uses 33 kV as its inverter-side voltage base, the prefault terminal RMS voltage is approximately 1 pu, as observed in Fig. 3.8.

Under the prefault condition, the inverter operates within the normal voltage range and the outer control loops are able to regulate the active and reactive powers close to their commanded references. With $V_{rms} \approx 1$ pu, the expected current magnitude can be obtained directly from the complex-power relationship,

$$I_{rms} = S/V_{rms} = \frac{\sqrt{P^2 + Q^2}}{V_{rms}} = \frac{\sqrt{0.8^2 + 0.2^2}}{1} \approx 0.82 \text{ pu} \quad (3.1)$$

This value agrees with the prefault current shown in Fig. 3.8. Similarly, Fig. 3.9 shows that after the initial start-up transient has settled, the measured active and reactive powers approach $P = 0.8$ pu and $Q = 0.2$ pu, confirming that the GFL controller is successfully tracking P_{ref} and Q_{ref} before the fault.

This initialization of the prefault controller state is important because P_{ref} , Q_{ref} , and the controller states established before the voltage depression subsequently influence the measured fault response. At $t = 1$ s, the voltage disturbance causes the inverter to leave this normal power-regulation condition. As shown in Fig. 3.8, the terminal voltage falls sharply from approximately 1 pu toward the low-voltage region, while the inverter current increases toward its 1.2-pu current limit. Fig. 3.9 simultaneously shows that the measured P and Q do not remain at their original references

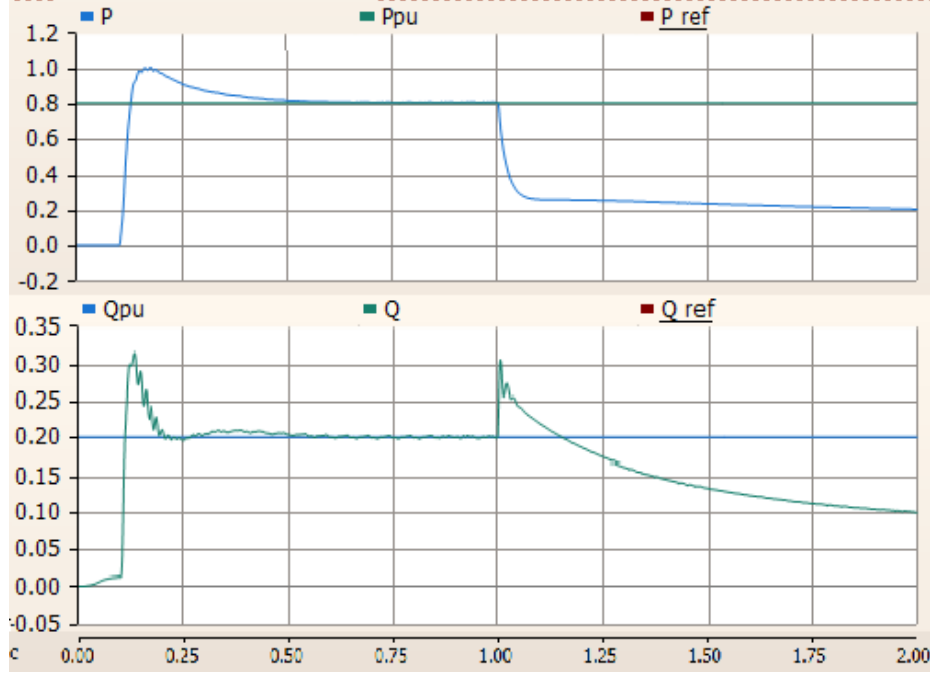


Figure 3.9: Time Domain Signals P and Q with Fault Introduced at $t=1s$. P and Q are measured from the LCL Filter and are subject to the slow response of the inverter outer control loop.

and instead decrease as the terminal voltage collapses. This transition from the well-defined prefault operating point to the low-voltage, current-constrained response is the behavior characterized by the empirical model developed in Section 5.

3.4.2 Fault Transient Data

Following the prefault verification, a series of simulations is performed by sweeping P_{ref} , Q_{ref} , terminal-voltage magnitude, prefault voltage, voltage phase, and the fault-inception time. For each case, the terminal voltage and current together with the measured active and reactive powers are recorded after the postfault response has sufficiently settled. The postfault quantities are averaged over this sustained portion of the waveform so that the resulting dataset characterizes the controlled inverter response instead of the instantaneous switching or filter transient immediately following fault inception. Figures 3.10–3.14 summarize the principal relationships obtained from these sweeps and form the dataset used subsequently to identify the dominant variables and construct the analytical model.

Figure 3.10 shows the measured terminal active power as a function of terminal-voltage magnitude for different values of P_{ref} . At voltages above approximately $V_{freeze} = 0.8$ pu, the active power approaches its respective reference value, corresponding to normal closed-loop operation of the GFL controller. Once the voltage falls below the freeze threshold, the controller cannot maintain the original active-power reference in the same manner, and P decreases progressively with terminal voltage. Figure 3.11 further shows that the postfault active power depends on the prefault voltage. For the same P_{ref} and approximately the same postfault terminal voltage, the cases initialized at $V_{prefault} = 0.9, 1.0$, and 1.1 pu have different active-power output results. This confirms that

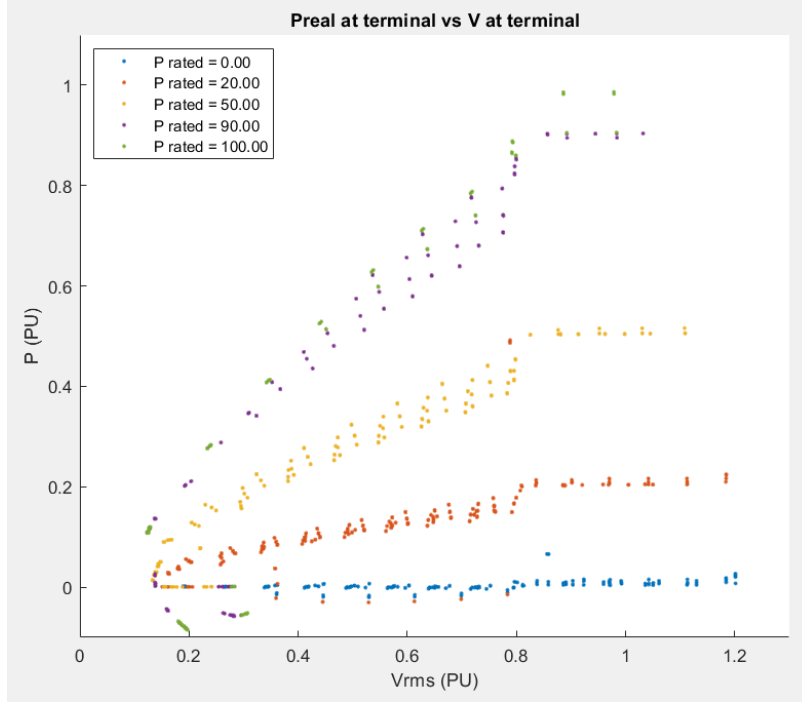


Figure 3.10: Measured Terminal Active Power P_{real} versus Terminal Voltage for Different P_{ref} .

the fault response retains information about the operating condition established before the voltage disturbance and motivates the inclusion of $V_{prefault}$ in the model. A similar voltage dependence is observed for reactive power in Fig. 3.12. Above the freeze threshold, the measured Q approaches the commanded Q_{ref} , whereas below 0.8 pu the individual trajectories become increasingly nonlinear. Unlike active power, the reactive-power response generally remains positive even at severe voltage depressions because reactive-current control is prioritized in the inverter configuration used in this study. The difference between the P and Q trajectories determines the phase of the current and the division between the d - and q -axis components for the current limit. The combined effect of P_{ref} and Q_{ref} on current magnitude is illustrated in Fig. 3.13 by grouping the cases according to $S_{ref} = \sqrt{P_{ref}^2 + Q_{ref}^2}$. In general, larger S_{ref} produces a larger current requirement, consistent with the approximate relationship $I \approx S/V$ before current saturation. As the terminal voltage decreases, the current required to maintain the commanded power increases until the converter-current constraint becomes dominant, as observed in the upper branches of Fig. 3.13.

Finally, Fig. 3.14 examines the current phase as a function of terminal voltage for different reactive-power operating points. At moderate and high voltage, the results form relatively distinct phase bands: cases with $Q_{ref} = 0$ remain near approximately -30° , while increasing Q_{ref} progressively rotates the current toward more negative angles, reaching approximately -120° for the highest reactive-power cases. This behavior is consistent with the dq -frame relationship $\theta_I - \theta_{PLL} \approx \text{atan2}(-Q, P)$.

In Figs. 3.10-3.14, at very low voltage, particularly below approximately 0.2–0.3 pu, the trajectories instead exhibit bending, branching, and partial convergence. The weaker terminal-voltage reference combined with the state-freeze behavior, current limitation, and reactive-current priority make the phase increasingly dependent on the retained controller and synchronization state rather

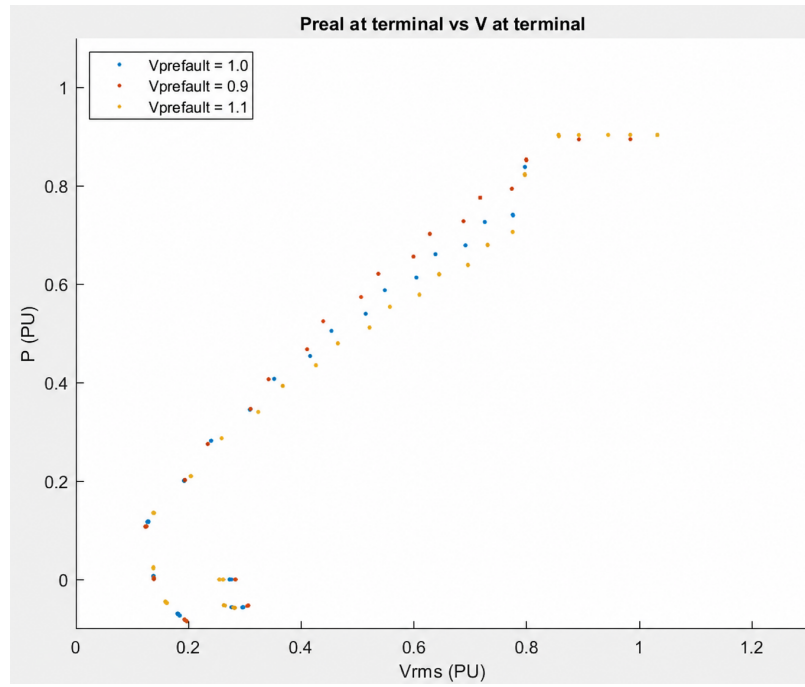


Figure 3.11: Measured Terminal Active Power P_{real} versus Terminal Voltage for $P_{ref} = 0.9$, Illustrating the Effect of Prefault Voltage.

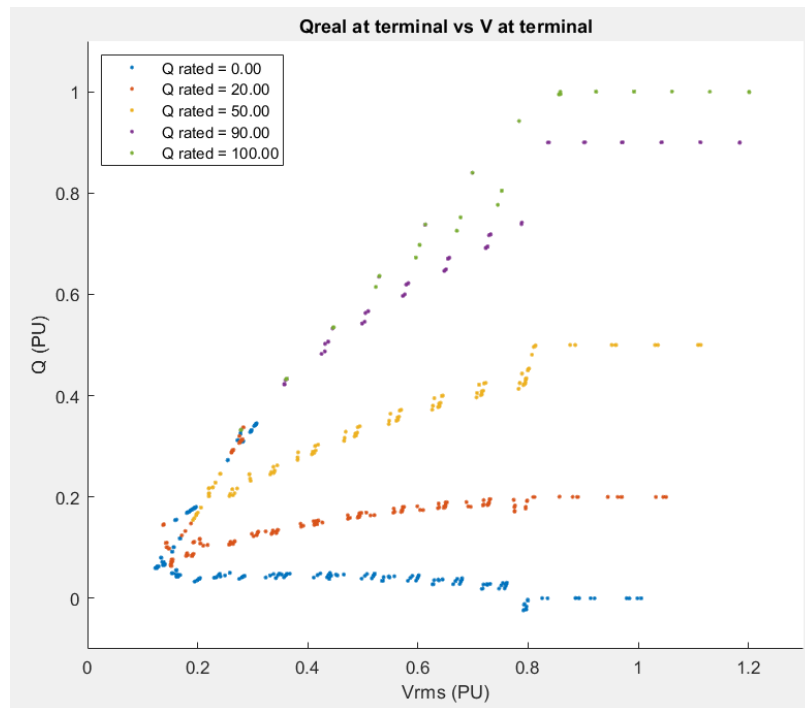


Figure 3.12: Measured Terminal Reactive Power Q_{real} versus Terminal Voltage for Different Q_{ref} .

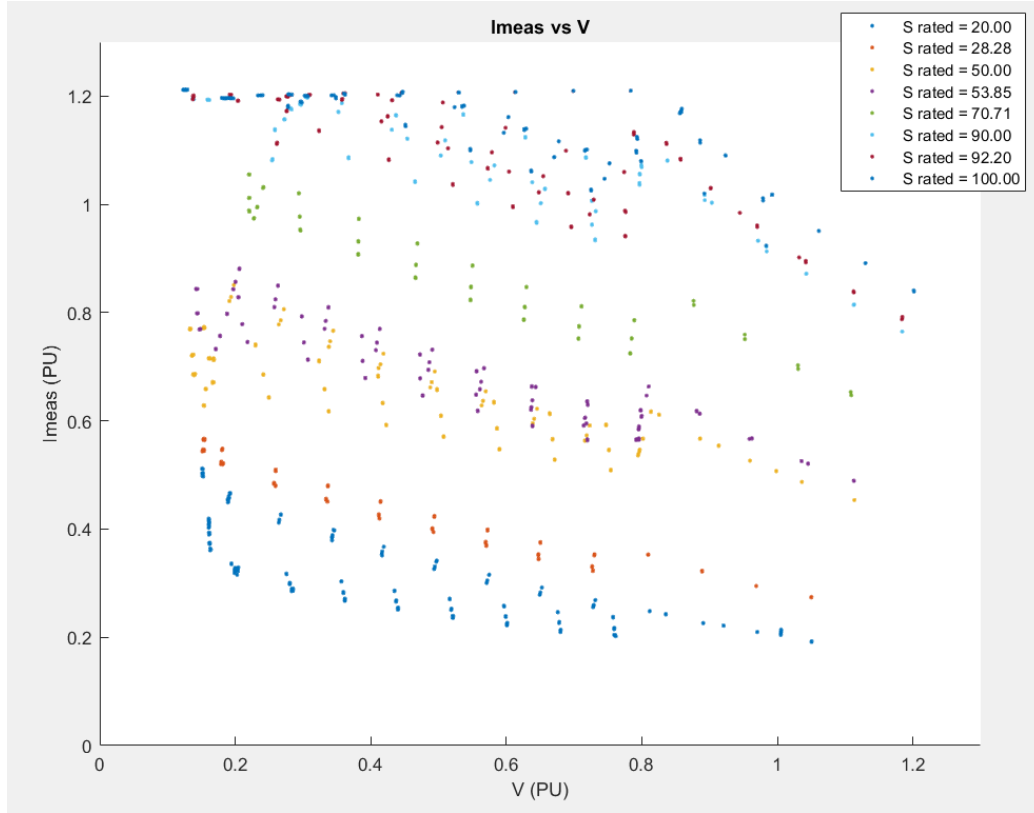


Figure 3.13: Measured Inverter Current Output I at Terminal versus Terminal Voltage for Different S_{ref} .

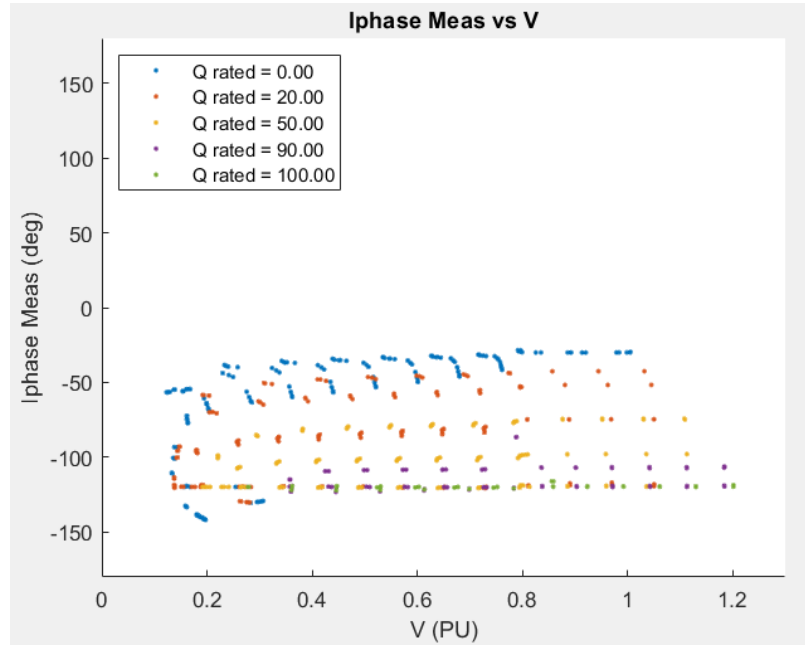


Figure 3.14: Measured Current Phase I_{phase} at Terminal versus Terminal Voltage for Different Q_{ref} .

than on the instantaneous P/Q ratio alone. A smaller change is also visible near $V_{\text{freeze}} \approx 0.8$ pu as the controller transitions between normally evolving and frozen states.

To prevent overfitting the model with the obtained PSCAD simulation data, the dataset is separated into a training dataset and a validation dataset. The majority of the data obtained from the PSCAD simulations are assigned to the training dataset to perform least-squares fitting for the model parameters. Selected P_{ref} and Q_{ref} values such as $P_{\text{ref}} = 0.8$ pu and $Q_{\text{ref}} = 0.3$ pu were withheld from the fitting dataset and used exclusively for validation. This method helps assess generalization. Using fewer fitted parameters makes the model less likely to reproduce small irregularities in the training data and more likely to capture the underlying physical trend. The final testing is done through the 3-bus network simulations (Fig. 3.7) to confirm the validity of the model itself.

Chapter 4

Physical Analysis of Fault Current

4.1 Effect of Voltage Magnitude

Figures 3.10-3.13 show that terminal-voltage magnitude is one of the dominant variables governing the fault response of the inverter, but its effect is not described by a single voltage-current relationship over the full operating range. Instead, the results exhibit a clear change in behavior around the low-voltage state-freeze threshold $V_{\text{freeze}} = 0.8$ pu. In the EPRI model used in this study, the inverter is operated as a GFL resource with an SRF-PLL, reactive-current priority is selected, and the converter current is limited to $I_{\text{max}} = 1.2$ pu. Figures 3.10-3.13 should be interpreted together: the voltage determines how much current is required to support the commanded P and Q , while the converter current limit and the low-voltage freeze logic determine whether those commands can actually be maintained.

For $V \geq V_{\text{freeze}}$, the behavior in Figures 3.10 and 3.12 is consistent with normal closed-loop $P - Q$ regulation. At higher terminal voltages, the curves become approximately horizontal at their respective reference values. Physically, when the PLL remains approximately aligned with the terminal voltage, $V_q \approx 0$, so the power relationships reduce to Equations (2.17) and (2.18) and maintaining the references requires

$$I_d \approx \frac{P_{\text{ref}}}{V}, \quad I_q \approx -\frac{Q_{\text{ref}}}{V} \quad (4.1)$$

Then, the required current magnitude is approximately

$$I_{\text{req}} = \sqrt{(I_d)^2 + (I_q)^2} \approx \frac{\sqrt{P_{\text{ref}}^2 + Q_{\text{ref}}^2}}{V} = \frac{S_{\text{ref}}}{V} \quad (4.2)$$

This explains the trend in Fig. 3.13: for a given S_{ref} , current increases as voltage decreases, while a larger S_{ref} produces a larger current at the same voltage. The relationship holds only until the requested current approaches I_{max} . The high- S_{ref} cases in Fig. 3.13 begin to approach the 1.2-pu ceiling as the voltage is reduced, showing that current limitation can occur even before the 0.8-pu freeze threshold is crossed: current saturation and state freezing are two separate nonlinear boundaries.

For $V < V_{\text{freeze}}$, Figures 3.10 and 3.12 show a qualitatively different response. Instead of continuing to hold $P \approx P_{\text{ref}}$ and $Q \approx Q_{\text{ref}}$, both powers become strongly voltage dependent and generally

decrease as the voltage is reduced. The model description specifies that below the 0.8-pu threshold, the controller does not attempt to maintain the original P_{ref} and Q_{ref} in the same manner and instead emphasizes voltage support and current limitation [1]. It should also be noted that the terminal voltage does not decrease completely to zero in the simulations. This is due to the minimum voltage-magnitude limit of approximately 0.2 pu imposed by the VSC phase-voltage reference controller, as discussed in Section 3.2.2; the simulated voltage response exhibits a lower-voltage floor near this value. The reduction in P and Q can also be understood directly from the converter current constraint. Since Equation (2.35) holds, the maximum transferable apparent power satisfies Equation (2.21). Once current is constrained, lowering voltage necessarily shrinks the available $P - Q$ capability even if the original references remain unchanged. This is consistent with Ravanji et al. [3], who show that the feasible converter $P - Q$ region contracts with decreasing terminal voltage.

Figure 3.13 provides the clearest summary of the transition. Above V_{freeze} , the current can be interpreted primarily through the constant-power tendency $I_{\text{req}} \approx S_{\text{ref}}/V$, followed by saturation when the requested current reaches I_{max} . Below V_{freeze} , the behavior of the distinct S_{ref} branches is determined by the controller state at entry into the frozen region, the remaining voltage-support logic, the Q -priority allocation, and the current limiter. The largest- S_{ref} cases are driven toward the 1.2-pu ceiling, whereas lower- S_{ref} cases remain below the limit and follow separate voltage-dependent trajectories. These observations are in line with the saturation regions found in Equation (2.60).

4.2 Effect of P Reference and Q Reference

The prefault active- and reactive-power references determine the initial dq -current requirements of the GFL inverter and have a direct influence on its subsequent fault-current response. Figures 3.10 and 3.12 demonstrate this dependence separately. At sufficiently high voltage, the measured active power in Fig. 3.10 approaches the specified P_{ref} , while the measured reactive power in Fig. 3.12 similarly approaches Q_{ref} . Figure 3.13 then combines these effects through S_{ref} and shows that cases with larger apparent-power references generally produce larger current magnitudes. Thus, before the converter encounters a control or current limit, P_{ref} and Q_{ref} influence the current magnitude primarily through their combined apparent-power requirement.

However, the individual values of P_{ref} and Q_{ref} remain important because they determine the orientation of the current vector. From (4.1),

$$\frac{I_q}{I_d} \approx -\frac{Q_{\text{ref}}}{P_{\text{ref}}} \quad (4.3)$$

so that the current angle relative to the PLL reference according to Equation (2.27) can be written approximately as

$$\theta_I - \theta_{\text{PLL}} \approx \tan^{-1} \left(\frac{I_q}{I_d} \right) \approx \tan^{-1} \left(-\frac{Q_{\text{ref}}}{P_{\text{ref}}} \right) \quad (4.4)$$

In Figures 3.13 and 3.14, because the PLL uses the terminal voltage as its phase reference, the relative current angle is largely determined by the Q/P ratio, which changes only modestly between the prefault and postfault operating points in many of the simulated cases. Two operating points with approximately the same S_{ref} can produce similar current magnitudes but different current phases if their $Q_{\text{ref}}/P_{\text{ref}}$ ratios differ. The same current magnitude does not imply the same P , Q ,

or apparent impedance.

The influence of P_{ref} and Q_{ref} changes once the requested current reaches the converter limit. In this EPRI model, $I_{\text{max}} = 1.2$ pu and reactive-current priority is selected. When

$$I_{\text{req}} = \sqrt{\left(\frac{P_{\text{ref}}}{V}\right)^2 + \left(\frac{Q_{\text{ref}}}{V}\right)^2} = \frac{\sqrt{P_{\text{ref}}^2 + Q_{\text{ref}}^2}}{V} > I_{\text{max}} \quad (4.5)$$

the inverter cannot maintain both references simultaneously. With Q -priority, the limiting process can be represented approximately by Equation (2.42). Increasing Q_{ref} can reduce the current capability remaining for active power once saturation occurs. In other words, P_{ref} and Q_{ref} are not symmetric once the limiter is active: reactive current is retained preferentially, while I_d , and P , may be curtailed. This helps explain why the high-reference cases in Figs. 3.10 and 3.12 depart from their commanded powers as voltage decreases even though the current in Fig. 3.13 remains bounded near I_{max} .

Finally, below $V_{\text{freeze}} = 0.8$ pu, P_{ref} and Q_{ref} cease to act as simple instantaneous targets. The implemented controller disables the relevant PI-integrator action when V_d leaves the permissible state-freeze range. This is visible in Figs. 3.10 and 3.12, where the different curves with their respective reference values remain clearly distinguishable but do not follow horizontal lines at their reference values below the freeze boundary. The references continue to matter because they determine the prefault I_d/I_q operating point and the controller state entering the fault.

4.3 Effect of Current Limit, State Freeze, and PLL Behavior

Current limiting is one of the dominant mechanisms shaping the inverter fault-current response because the converter cannot increase its current indefinitely as terminal voltage decreases. Equation (2.21) is particularly important physically. This behavior is visible when Figures 3.10-3.13 are considered together. Figure 3.13 shows the higher- S_{ref} cases clustering near the approximately 1.2-pu current ceiling, whereas Figs. 3.10 and 3.12 show that the corresponding P and Q do not remain at their original reference values as voltage is reduced to maintain either the voltage conditions or the current limit.

The low-voltage state-freeze mechanism introduces an important history dependence into the inverter fault response. In the EPRI model used in this thesis, the lower freeze threshold is $V_{\text{freeze}} = 0.8$ pu. When V_d falls below the threshold, the relevant PI-integrator states stop evolving normally and retain their previous values. The postfault response is determined by the instantaneous values of V , P_{ref} , and Q_{ref} and also by the controller state established before the threshold was crossed. This transition is visible in Figs. 3.10 and 3.12: above approximately 0.8 pu, P and Q remain close to their respective references, whereas below the threshold the trajectories become strongly voltage dependent and do not follow constant-reference behavior. Figure 3.11 further shows that different prefault voltages can produce different postfault active powers at similar fault voltages, illustrating the memory introduced by the frozen controller states.

Chapter 5

Development of the Empirical Fault Calculation Model

This chapter develops the reduced-order empirical model from the physical trends identified in the PSCAD simulations. The formulation begins with the fundamental positive-sequence dq -frame relationships that describe normal GFL inverter operation, then introduces corrections only where the EMT results show that those relationships are not sufficient. The objective is to reproduce the dominant sustained fault response using a compact set of algebraic equations.

5.1 Base Physical Model

The model is first constructed from the fundamental positive-sequence power relationships of a GFL inverter. For the balanced faults considered in this work, the PLL defines the local dq reference frame and, when synchronization is maintained, $\theta_{\text{PLL}} \approx \theta_V$, such that $V_d \approx V$ and $V_q \approx 0$. Under this condition, the active and reactive powers are approximately

$$P = VI_d, \quad Q = -VI_q \quad (5.1)$$

which gives

$$I_d = \frac{P}{V}, \quad I_q = -\frac{Q}{V} \quad (5.2)$$

The corresponding current magnitude and current angle relative to the terminal voltage are

$$I = \sqrt{I_d^2 + I_q^2} = \frac{\sqrt{P^2 + Q^2}}{V} \quad (5.3)$$

and with Equation (2.28) and the assumed PLL angle of the inverter aligned with the voltage phase

$$\theta_I \approx \theta_{\text{PLL}} + \tan^{-1} \left(\frac{I_q}{I_d} \right) = \theta_{\text{PLL}} + \text{atan2}(-Q, P) \quad (5.4)$$

These relationships provide the physical starting point for the empirical model: P primarily determines the d -axis current, Q determines the q -axis current, their vector magnitude determines the total current, and their ratio determines its orientation. This is consistent with the simulation

results in Section 3.4, where S_{ref} largely controls current magnitude while the P/Q ratio strongly influences current orientation. These fundamental relationships also provide a good approximation of the inverter during normal, unfaulted operation, when the PLL remains well synchronized, $V_q \approx 0$, and the controller is not constrained by low-voltage state freezing. Under these conditions, the outer control loops regulate $P \approx P_{\text{ref}}$ and $Q \approx Q_{\text{ref}}$, so the resulting I_d , I_q , current magnitude, and current phase can be estimated accurately from the basic dq -frame equations before any empirical correction is required.

5.2 Active Power Model

The starting point for the active-power model is the physical dq -frame relationship developed in Section 5.1. When the PLL is approximately aligned with the terminal voltage, $P \approx VI_d$. Above the low-voltage freeze threshold, the outer controller can continuously modify I_d in an attempt to maintain $P \approx P_{\text{ref}}$. Below $V_{\text{freeze}} = 0.8$ pu, this assumption no longer holds. The EPRI controller freezes the relevant PI-integrator state when V_d leaves the allowed voltage range, so the active-current reference is not continuously adjusted to compensate for the decreasing terminal voltage. This behavior is visible in Fig. 3.10: the different P_{ref} cases remain separated below V_{freeze} , showing that the prefault operating point determines the scale of the response. This observation suggested a model in which a prefault active-power scale is multiplied by a voltage-dependent reduction term.

If I_d were completely constant after the controller entered the frozen region, the basic relation $P = VI_d$ would predict a linear voltage dependence,

$$\frac{P}{P_{\text{ref}}} \approx \left(\frac{V}{V_{\text{prefault}}} \right)^1 \quad (5.5)$$

where P_{ref} is also the prefault P , V_{prefault} is the prefault voltage, and the exponent of 1 indicates the linear relationship. Perfect constant-power regulation would correspond to essentially no reduction of P with voltage, or an exponent of zero. The PSCAD results in Fig. 3.10 fall between these two limiting behaviors: active power clearly decreases as voltage is reduced, but its dependence is not exactly linear. This is physically reasonable because I_d is not simply fixed; below the freeze threshold, its value is also affected by the frozen controller state, current limitation, and the reactive-current-priority logic. In particular, the converter must satisfy Equation (2.35) and (2.42). A generalized power law is introduced,

$$P_{\text{calc}} = P_{\text{ref}} \left(\frac{V}{V_{\text{prefault}}} \right)^\alpha, \quad V < V_{\text{freeze}} \quad (5.6)$$

where α allows the empirical model to interpolate between constant-power and constant-current-like behavior. Fitting this relationship to the low-voltage PSCAD data showed that an exponent of approximately

$$\alpha = 0.8 \quad (5.7)$$

provided a suitable representation, giving the implemented expression

$$P_{\text{calc}} = \begin{cases} P_{\text{ref}} \left(\frac{V_{1,\text{inv}}}{V_{\text{prefault}}} \right)^{0.8} & , V_{1,\text{inv}} < V_{\text{freeze}} \\ P_{\text{ref}} & , V_{1,\text{inv}} \geq V_{\text{freeze}} \end{cases} \quad (5.8)$$

The exponent 0.8 should be interpreted as an empirical correction to the ideal $P \propto V$ relationship, rather than as a fundamental converter constant. Its value being less than one is consistent with the observation that the inverter retains some active-current support as voltage decreases, so active power falls more slowly than would be predicted by a strictly frozen I_d . At the same time, $\alpha > 0$ ensures that the model departs from constant-power operation and forces $P_{\text{calc}} \rightarrow 0$ as $V \rightarrow 0$, which is physically consistent with the finite-current constraint Equation (2.21).

Finally, normalizing the postfault voltage by V_{prefault} is important because it expresses the severity of the voltage depression relative to the inverter's initial operating condition rather than as an absolute voltage. When $V = V_{\text{prefault}}$, the voltage factor becomes 1, and the equation returns $P_{\text{calc}} = P_{\text{ref}}$; as the fault voltage decreases, the normalized term smoothly reduces the calculated active power. This formulation preserves the two dominant trends observed in Fig. 3.10: the initial active-power operating point sets the vertical scale, while the normalized voltage depression determines the degree of active-power reduction below the freeze threshold. The 0.8 exponent then captures, in a single compact parameter, the departure from the ideal linear $P = VI_d$ behavior caused by the nonlinear converter controls observed in PSCAD.

5.3 Reactive Power Model

The reactive-power model begins from the same synchronized dq -frame relationship used in Section 5.1. Above V_{freeze} , the control loop can continually adjust I_q so that Q remains close to Q_{ref} . Below $V_{\text{freeze}} = 0.8$ pu, the relevant PI-integrator state is frozen, and the inverter does not regulate the original P_{ref} and Q_{ref} in the same manner. Figure 3.12 reflects this transition: the different Q_{ref} cases remain clearly separated, showing the effect of the prefault reactive-power operating point.

Below V_{freeze} , the frozen controller state motivates a voltage-dependent reactive-power relationship based on the inverter terminal voltage; an initial candidate for the nonlinear voltage dependence was the familiar voltage-source-behind-reactance relationship from Equation (2.10)

$$Q \approx \frac{V(E - V)}{X} \quad (5.9)$$

which is closely related to the conventional reactive-power transfer equation for a voltage source behind a predominantly inductive impedance [13]. Using the grid-side sign convention adopted here and the small-angle approximation, the voltage-source-behind-reactance relationship from Equation (2.10) reduces to (5.9). It naturally produces a curved $Q - V$ characteristic and forces $Q \rightarrow 0$ as $V \rightarrow 0$. However, the model is operated as a GFL controlled-current source, instead of a fixed voltage source behind a constant impedance. A single constant E and X cannot represent all of the Q_{ref} cases in Fig. 3.12. In particular, (5.9) contains no explicit information about the prefault reactive-power command, even though the simulations show that this command continues to separate the different low-voltage trajectories.

Nevertheless, Equation (5.9) provides a useful mathematical starting point if it is anchored to

the prefault operating condition. Suppose the prefault reactive power is Q_{ref} at $V_{prefault}$. Applying (5.9) at the prefault operating point gives

$$Q_{ref} = \frac{V_{prefault}(E - V_{prefault})}{X} \quad (5.10)$$

from which

$$E = V_{prefault} + \frac{XQ_{ref}}{V_{prefault}} \quad (5.11)$$

Substituting this result back into (5.9) eliminates the unknown internal voltage E and gives

$$Q = \frac{V}{V_{prefault}}Q_{ref} + \frac{V}{X}(V_{prefault} - V) \quad (5.12)$$

Equation (5.12) separates the response into two physically interpretable contributions: a term representing the reactive power associated with the prefault q -axis current, and a second term representing the additional response to the voltage depression. This structure is much closer to the behavior observed in Fig. 3.12 than the original $V(E - V)/X$ expression. The first term in the implemented model,

$$Q_{base} = \frac{V}{V_{ref}}Q_{ref} \quad (5.13)$$

can also be obtained directly from the GFL current-control interpretation, where V_{ref} is the per-unit voltage reference of the inverter. In the implementation, V_{ref} is used for this current normalization instead of $V_{prefault}$ because it is assumed that the Q - V control is able to match $V_{prefault}$. Immediately before the low-voltage state is entered,

$$I_{q0} \approx -\frac{Q_{ref}}{V_{ref}} \quad (5.14)$$

and if the frozen controller causes this current component to vary less than it would under normal closed-loop Q regulation, then

$$Q \approx -VI_{q0} = \frac{V}{V_{ref}}Q_{ref} \quad (5.15)$$

Thus, the first term represents a frozen-current baseline: it retains the prefault reactive-power operating point but allows the delivered reactive power to fall as terminal voltage collapses. The remaining error after this baseline term is not constant across the different Q_{ref} cases, indicating that a single equivalent $1/X$ is insufficient. The second term of (5.12) is effectively a virtual voltage source behind a variable impedance depending on Q_{ref} , representing the combined low-voltage reactive response of the current controller, frozen state, current limiter, and reactive-current-priority logic. The present inverter explicitly prioritizes reactive current, so the response itself changes with the amount of reactive current already being requested. The residual PSCAD behavior was represented using an effective reactive-voltage coefficient k_Q that is fine-tuned using the empirical data obtained in Section 3.4 instead of a constant $1/X$. In this thesis, a linear relationship with Q_{ref} was used to determine k_Q :

$$k_Q = 0.25 + 0.175 Q_{ref} \quad (5.16)$$

The linear dependence on Q_{ref} provides the simplest correction capable of increasing the voltage-support term as the initial reactive-power operating point increases. The numerical coefficients

0.25 and 0.175 are empirically identified coefficients from the PSCAD low-voltage data and are not physical reactance parameters. The resulting low-voltage reactive-power model is

$$Q_{\text{calc}} = \begin{cases} \frac{V}{V_{\text{ref}}} Q_{\text{ref}} + k_Q V (V_{\text{prefault}} - V) & , V_{1,\text{inv}} < V_{\text{freeze}} \\ Q_{\text{ref}} & , V_{1,\text{inv}} \geq V_{\text{freeze}} \end{cases} \quad (5.17)$$

with

$$k_Q = 0.25 + 0.175 Q_{\text{ref}} \quad (5.18)$$

The two terms have complementary roles. The first preserves the dependence on the prefault Q operating point and approximates the effect of a retained q -axis current, while the second reproduces the curvature associated with the magnitude of the voltage depression. Importantly, the correction term is zero when $V = V_{\text{prefault}}$ and also disappears as $V \rightarrow 0$.

5.4 Current Magnitude Model

Once the active- and reactive-power models were established, the current-magnitude model follows directly from the positive-sequence complex-power relationship instead of requiring an independent empirical fit. For the balanced three-phase faults considered in this thesis,

$$S = P + jQ = VI^* \quad |S| = \sqrt{P^2 + Q^2} = VI \quad (5.19)$$

Solving for current magnitude gives

$$I_{\text{calc}} = \frac{\sqrt{P_{\text{calc}}^2 + Q_{\text{calc}}^2}}{V} \quad (5.20)$$

This formulation is consistent with the physical dq -frame model developed previously, where $P \approx VI_d$, $Q \approx -VI_q$, and hence $I = \sqrt{I_d^2 + I_q^2} = \sqrt{P^2 + Q^2}/V$. The same relationship was already observed in the PSCAD results of Fig. 3.13, where increasing $S_{\text{ref}} = \sqrt{P_{\text{ref}}^2 + Q_{\text{ref}}^2}$ increased the converter current for a given terminal voltage.

The important difference from the conventional constant-power expression is that (5.20) uses the calculated postfault powers, P_{calc} and Q_{calc} , rather than simply substituting P_{ref} and Q_{ref} . Below V_{freeze} , both P and Q depart from their references and become voltage dependent because the controller does not maintain the original P_{ref} and Q_{ref} in the same way. The voltage scaling in (5.20) should not be interpreted simply as $I \propto 1/V$. The numerator is itself a function of voltage through the models in Equations (5.8) and (5.17). This allows the derived current model to reproduce the nonlinear voltage dependence observed below V_{freeze} while retaining the fundamental power-current relationship.

A second essential step is the enforcement of the physical converter current limit. The unconstrained result from (5.20) is first evaluated and then compared with the maximum converter current. For the inverter studied here, $I_{\text{max}} = 1.2$ pu. Any value exceeding this limit is replaced by I_{max} ,

$$I_{\text{calc}} = \min \left[\frac{\sqrt{P_{\text{calc}}^2 + Q_{\text{calc}}^2}}{V}, I_{\text{max}} \right] \quad (5.21)$$

The saturation is required by the converter constraint Equation (2.35) and is strongly supported by Fig. 3.13, where the higher- S_{ref} cases converge toward the approximately 1.2-pu current ceiling as voltage decreases. Prior to saturation, a reduction in voltage tends to increase current because a greater current is required to transfer the calculated apparent power. Further voltage reduction cannot produce a corresponding increase in sustained current. Instead, the allowable apparent power contracts. This reproduces the two regimes identified in this study: an unsaturated region in which current varies with voltage and power demand, followed by a saturated region in which the current magnitude is fixed. The 1.2-pu current boundary and $V_{\text{freeze}} = 0.8$ pu state-freeze boundary are distinct; high-power cases can reach I_{max} before the voltage crosses V_{freeze} .

The final current-magnitude model preserves a useful separation between empirical and physical modeling. The nonlinear effects of voltage depression, prefault operating condition, state freezing, and reactive-current support are captured through P_{calc} and Q_{calc} , while the conversion from power to current remains governed by the exact phasor identity and the physical converter limit:

$$I_{\text{calc}} = \min \left(\frac{\sqrt{P_{\text{calc}}^2 + Q_{\text{calc}}^2}}{V_{1,\text{inv}}}, I_{\text{max}} \right) \quad (5.22)$$

This is preferable to fitting current independently because it maintains consistency between the calculated P , Q , and I for non-current-limiting data, while also reproducing the saturation observed in the PSCAD simulations at I_{max} . As mentioned before, $V \approx 0$ does not arise in the simulations considered here because the VSC phase-voltage reference controller imposes a minimum voltage magnitude of approximately 0.2 pu (Section 3.2.2). As a result, the terminal voltage does not collapse to zero, and the division by V in (5.22) remains well defined over the operating range. Physically, P and Q would approach zero as $V \rightarrow 0$. However, as this limiting condition is not reached in the present simulations, once the controller state is frozen, the retained q -axis current together with the nonzero terminal voltage allows a finite reactive-power output to remain, so Q does not collapse to zero in the modeled operating range.

5.5 Current Phase Model

The current-phase model was derived from the same positive-sequence dq -frame relationships used for the current-magnitude model. When the PLL is synchronized with the terminal voltage,

$$I_d = \frac{P}{V}, \quad I_q = -\frac{Q}{V} \quad (5.23)$$

and the orientation of the current vector in the PLL reference frame is

$$\theta_{dq} = \text{atan2}(I_q, I_d) \quad (5.24)$$

Substituting the power relationships gives

$$\theta_{dq} = \text{atan2} \left(-\frac{Q}{V}, \frac{P}{V} \right) = -\text{atan2}(Q, P), \quad V > 0 \quad (5.25)$$

Thus, although the current magnitude depends strongly on voltage through $I = \sqrt{P^2 + Q^2}/V$, the current orientation is primarily determined by the ratio between P and Q because the common voltage factor cancels. From the PSCAD observations in Fig. 3.14, the current phase separates mainly according to the active- and reactive-power operating point, while voltage has comparatively little direct influence over the intermediate-voltage region. The results similarly show that increasing Q shifts the current phase toward increasingly negative angles and that the phase is largely governed by the Q/P allocation when the PLL remains synchronized. Combining Equations (2.27) and (5.25) gives

$$\theta_I = \theta_V - \text{atan2}(Q, P) + \theta_{\text{off}} \quad (5.26)$$

where θ_V represents the terminal-voltage reference angle and θ_{off} accounts for the particular current-voltage phase offset. For the PSCAD measurements used to construct the empirical model, the terminal-voltage reference is already incorporated into the phase measurement ($\theta_{\text{PLL}} \approx \theta_V \approx 0$), leaving a nearly constant $\theta_{\text{off}} \approx -30^\circ$. From Fig. 3.14, when $Q_{\text{ref}} = 0$, the measured current phase approaches approximately -30° over the normally synchronized voltage region. $\theta_{\text{off}} = -\frac{\pi}{6}$ and Equation (5.26) reduces to

$$\theta_{I,\text{calc}} = -\text{atan2}(Q_{\text{calc}}, P_{\text{calc}}) - \frac{\pi}{6} \quad (5.27)$$

Voltage has not actually disappeared from the overall phase model. Rather, its effect remains indirectly through the voltage-dependent models $P_{\text{calc}}(V)$ and $Q_{\text{calc}}(V)$. This explains why the phase curves in Fig. 3.14 are comparatively flat at moderate voltage yet become nonlinear near V_{freeze} and $V \approx 0$, where P , Q , PLL synchronization, and frozen controller states themselves become strongly voltage dependent. A special treatment is required when $P_{\text{calc}} = 0$. For $P = 0$ and $Q > 0$,

$$-\text{atan2}(Q, 0) = -\frac{\pi}{2} \quad (5.28)$$

and

$$\theta_I = -\frac{\pi}{2} - \frac{\pi}{6} = -\frac{2\pi}{3} = -120^\circ \quad (5.29)$$

The -120° result is also consistent with Fig. 3.14, where the highly reactive or very-low-voltage cases converge toward approximately -120° . The complete empirical phase model can be summarized as

$$\theta_{I,\text{calc}} = \begin{cases} -\frac{2\pi}{3}, & P_{\text{calc}} = 0, Q_{\text{calc}} > 0, \\ -\text{atan2}(Q_{\text{calc}}, P_{\text{calc}}) - \frac{\pi}{6}, & P_{\text{calc}} \neq 0. \end{cases} \quad (5.30)$$

with $Q = 0$, $P > 0$ naturally yielding $-\pi/6$.

5.6 Parameter Identification

The parameters used in the empirical P_{calc} and Q_{calc} models were identified from the PSCAD simulation database in Section 3.4 using a nonlinear parameter-fitting procedure in MATLAB. The fitting data were formed by combining three voltage-drop datasets corresponding to different prefault-voltage conditions. Only cases satisfying

$$V \leq V_{\text{freeze}} = 0.8 \text{ pu} \quad (5.31)$$

are considered for the low-voltage parameter identification, because the empirical coefficients α and k_Q are intended specifically to describe the nonlinear frozen-controller region identified in Section 4.3.

The assumed physical forms of the active- and reactive-power equations were fixed first, and only a small number of coefficients were left for numerical identification. For active power, the candidate expression was

$$P_{\text{calc}} = P_{\text{ref}} \left(\frac{V}{V_{\text{prefault}}} \right)^\alpha \quad (5.32)$$

so the exponent α was represented directly by an optimization parameter. For reactive power, the proposed form was

$$Q_{\text{calc}} = \frac{V}{V_{\text{ref}}} Q_{\text{ref}} + V [(k_{Q0} + k_{Q1} Q_{\text{ref}}) (V_{\text{prefault}} - V)] \quad (5.33)$$

as

$$k_Q = k_{Q0} + k_{Q1} Q_{\text{ref}} \quad (5.34)$$

The MATLAB functions use α , k_{Q0} , and k_{Q1} as the fitting parameters. This approach is deliberately low-order: instead of fitting a large black-box function of voltage, P , and Q , the optimization adjusts only coefficients that already have a physically motivated role in the model equations. Limiting the number of fitted parameters also reduces the risk of overfitting the PSCAD dataset, helping the resulting model preserve the underlying physical trends and generalize more reliably to operating points that were not used directly in the parameter-identification process.

Before the parameters were combined into a single current-based optimization, the active- and reactive-power models were first fitted separately using the same general parameter-estimation procedure. For the active-power model, the assumed structure was (5.8) and α was varied to minimize the squared error between the calculated and measured PSCAD active power. Similarly, for reactive power, the candidate model (5.17) was fitted against the measured PSCAD reactive-power response, with k_{Q0} and k_{Q1} treated as unknown coefficients. In both cases, the optimization objective can be written generically as

$$J_P(\alpha) = \sum_i (P_{\text{meas},i} - P_{\text{calc},i})^2 \quad (5.35)$$

and

$$J_Q(k_{Q0}, k_{Q1}) = \sum_i (Q_{\text{meas},i} - Q_{\text{calc},i})^2 \quad (5.36)$$

The separately obtained coefficients were used as the starting point for the joint current-based optimization. The purpose of the subsequent joint optimization was not to construct the P and Q models from scratch, but to refine and verify the previously fitted coefficients in combination, using the final short-circuit current as the common output criterion. In this way, the fitting process was hierarchical: first the individual power relationships were calibrated, and then their interaction was assessed through the current equation.

The calculated powers were then converted into the inverter current magnitude:

$$I_{\text{model}} = \min \left[I_{\text{max}}, \frac{\sqrt{P_{\text{calc}}^2 + Q_{\text{calc}}^2}}{V} \right] \quad (5.37)$$

The advantage of fitting through (5.37) is that α , k_{Q0} , and k_{Q1} are identified jointly according to their effect on the final fault-current prediction. For example, changing α changes P_{calc} , while changing k_Q changes Q_{calc} , but both ultimately influence the same apparent-power magnitude and may interact when the 1.2-pu current ceiling is reached. This is more appropriate for the present objective than separately minimizing errors in P and Q , since an excellent fit to one power component could still give an inaccurate short-circuit current.

The optimization criterion is based on a least-squares error between the measured PSCAD current and the calculated current:

$$J(\mathbf{x}) = \sum_i (I_{\text{meas},i} - I_{\text{model},i})^2 \quad (5.38)$$

The final implementation uses

$$J(\mathbf{x}) = \sum_i \min((I_{\text{meas},i} - I_{\text{model},i})^2, 0.1) \quad (5.39)$$

instead. Equation (5.39) can be regarded as a capped least-squares criterion. For ordinary points with small errors, it behaves exactly like squared-error fitting, but an individual simulation case with a very large residual cannot dominate the entire optimization because its contribution is limited to 0.1. This is useful for the present dataset because the behavior near very low voltage and controller-transition boundaries is highly nonlinear and can contain cases that are difficult for a compact model to reproduce exactly.

MATLAB's *fmincon* routine is used to minimize this objective. The optimization is allowed up to 10^5 function evaluations and 5000 iterations. The final coefficients retained and verified/refined through the optimization procedure are as follows

$$\alpha = 0.8 \quad (5.40)$$

and

$$k_Q = 0.25 + 0.175 Q_{\text{ref}} \quad (5.41)$$

Chapter 6

Model Validation

The empirical validation criterion adopted for the proposed model is that the calculated inverter-current magnitude must remain within 0.1 pu of the corresponding PSCAD measurement. For each simulation case, a prediction is classified as a success when

$$|I_{\text{meas}} - I_{\text{calc}}| \leq 0.1 \text{ pu} \quad (6.1)$$

and the overall failure percentage is calculated as the number of failed cases divided by the total number of cases. In addition to this threshold-based metric, the model accuracy is quantified using the mean-squared error (MSE) for active power, reactive power, and current magnitude,

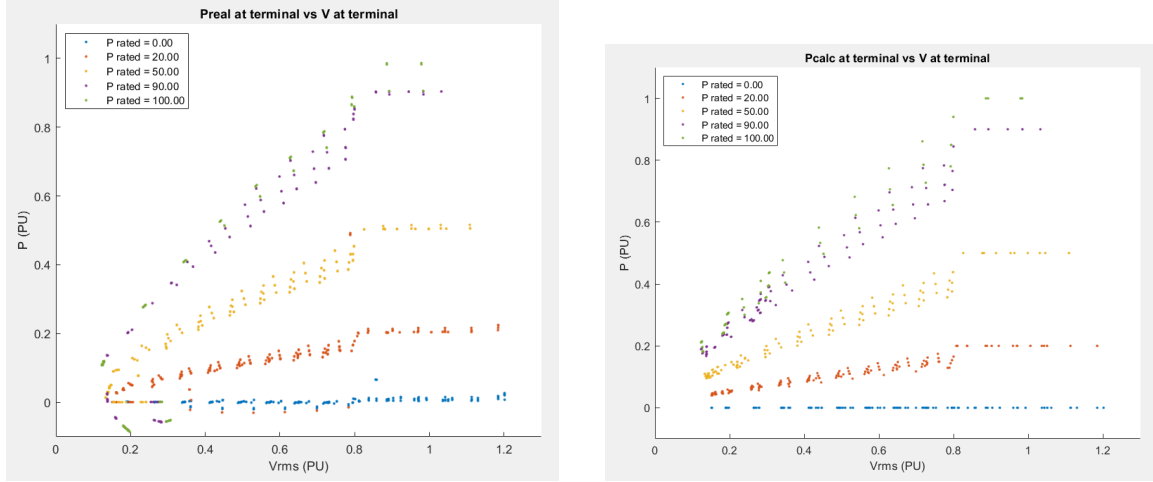
$$\text{MSE}_P = \frac{1}{N} \sum_{i=1}^N (P_{\text{meas},i} - P_{\text{calc},i})^2 \quad (6.2)$$

$$\text{MSE}_Q = \frac{1}{N} \sum_{i=1}^N (Q_{\text{meas},i} - Q_{\text{calc},i})^2 \quad (6.3)$$

$$\text{MSE}_I = \frac{1}{N} \sum_{i=1}^N (I_{\text{meas},i} - I_{\text{calc},i})^2 \quad (6.4)$$

while the current-phase accuracy is evaluated using the root-mean-square error.

Validation is performed progressively to determine whether the empirical relationships remain accurate beyond the cases used for parameter identification. The model is first assessed against the PSCAD simulation dataset using the above error measures and the 0.1-pu current-magnitude requirement. It is then applied to a larger network containing three buses and two IBRs in order to evaluate whether the proposed model remains applicable when multiple IBRs interact through the network rather than being studied individually (Fig. 3.7). Finally, a convergence test is performed to examine the numerical behavior of the proposed calculation procedure and determine whether the iterative network solution converges reliably when the empirical inverter model is incorporated into the short-circuit calculation.



(a) Measured Terminal Active Power P_{real} vs Terminal Voltage for Different P_{ref} .

(b) Calculated Terminal Active Power P_{model} vs Terminal Voltage for Different P_{ref} .

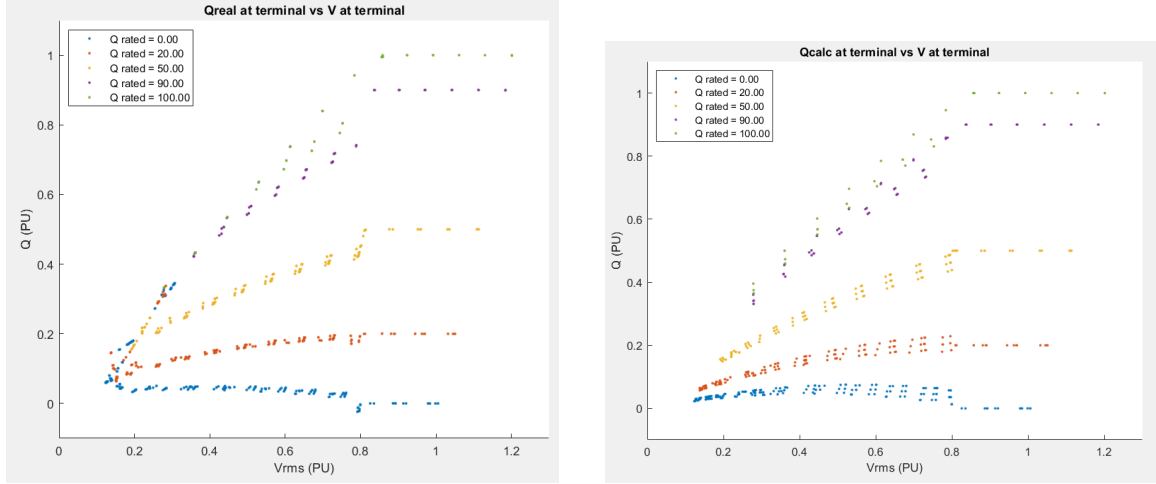
Figure 6.1: Comparison between the measured P at the inverter terminal in (a) and the predicted P by the model in (b).

6.1 P , Q , I Validations

Comparison of the PSCAD measurement with the results in Figures 6.1-6.4 shows that the proposed model reproduces the principal voltage-dependent structure of the inverter response while smoothing some of the smaller nonlinear variations produced by the detailed controller. For active power, Fig. 6.1 shows that the measured P values separate according to P_{ref} , approach their respective references at higher voltage, and decrease progressively once the terminal voltage falls below the low-voltage operating region. The calculated response in Fig. 6.1 reproduces the same ordering and overall curvature. The agreement indicates that the exponent $\alpha = 0.8$ captures the departure from both ideal constant-power behavior and the strictly linear $P \propto V$ response. The curves derived from the model are nevertheless more regular than the measurements from the EMT model, particularly in the lowest-voltage region.

A similar situation is observed between the measured reactive power and the calculated reactive power in Fig. 6.2. The PSCAD data show distinct Q_{ref} families and a nonlinear reduction of reactive power with decreasing terminal voltage, with the strongest curvature occurring below V_{freeze} . Figure 6.2 reproduces both the separation between the different reactive-power operating points and the curved low-voltage behavior. Visually, the modeled Q - V trajectories closely reproduce the broad shape of the measured trajectories in Figure 6.2, including the increasing reactive response associated with larger Q_{ref} . The remaining differences are concentrated mainly around the low-voltage and controller-transition regions, where the EMT model retains PI-integrator state, current-priority interactions, and PLL dynamics that are not individually represented in the compact Q equation.

The comparison in Figure 6.3 is particularly important because current magnitude is the primary validation quantity of the proposed short-circuit model. Figure 3.13 shows that the measured current increases with $S_{ref} = \sqrt{P_{ref}^2 + Q_{ref}^2}$, increases as voltage is depressed in the unsaturated region, and becomes bounded by the $I_{max} = 1.2$ pu converter limit. The model reproduces these same



(a) Measured Terminal Reactive Power Q_{real} vs Terminal Voltage for Different Q_{ref} .

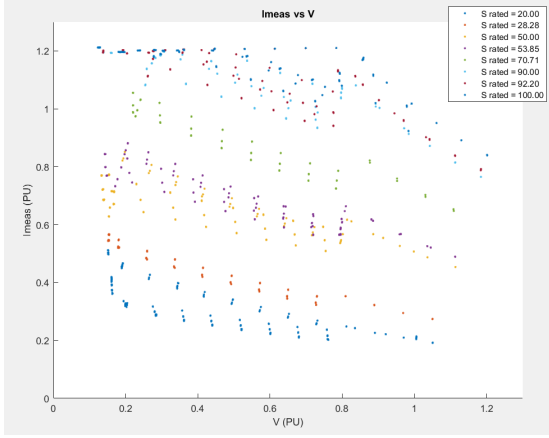
(b) Calculated Terminal Reactive Power Q_{model} vs Terminal Voltage for Different Q_{ref} .

Figure 6.2: Comparison between the measured Q at the inverter terminal in (a) and the predicted Q by the model in (b).

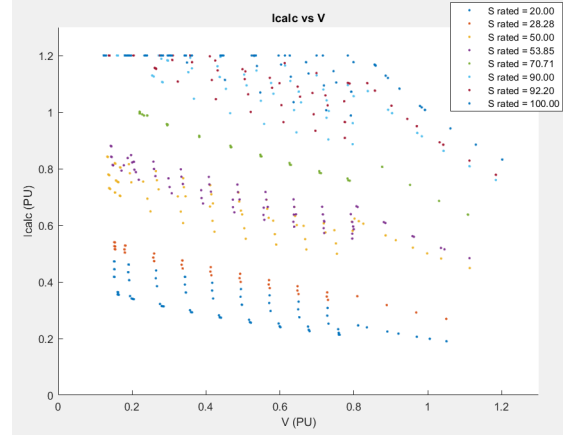
features: the different S_{ref} families remain ordered in approximately the same manner, the low- and intermediate-voltage slopes are retained, and the high-current cases terminate at the same 1.2-pu ceiling.

The current-phase comparison in Figure 6.4 shows both the strength and the main limitation of the reduced model. The measured results contain clear phase bands controlled primarily by the Q/P ratio: the $Q_{ref} = 0$ cases approach approximately -30° , while increasingly reactive operating points rotate the current toward approximately -120° . The model reproduces these principal bands through $\theta_{I,calc} = -\text{atan2}(Q_{calc}, P_{calc}) - \pi/6$, including the -30° limit for $Q = 0$ and the -120° limit as $P \rightarrow 0$. However, the measured phase exhibits more bending, branching, and scatter near $V \approx 0$, as well as smaller changes around V_{freeze} , whereas the phase calculated by the model is more regular.

The validation results show that the proposed model satisfies Equation (6.1) for the large majority of the PSCAD database. Only 1.05% of the simulated operating points are classified as failures. This failure rate decreases rapidly when the allowable error band is relaxed only slightly: increasing the tolerance to 0.11 pu reduces the failure percentage to approximately 0.50%, while a 0.12-pu tolerance reduces it further to 0.18%. Importantly, the remaining large-error cases are not distributed uniformly throughout the operating space. They occur primarily at extreme prefault conditions in which either P_{ref} or Q_{ref} is very large, typically 0.9–1.0 pu, while the other component is approximately zero. These corner cases correspond to highly asymmetric d/q -current allocations and represent some of the most demanding operating conditions for the reduced model. In addition, when the fault introduces a large terminal-voltage phase shift, the errors generally increase as the magnitude of the imposed phase shift increases in the tested cases. In these cases, the largest errors are concentrated mainly near the low-voltage operating limit and around the V_{freeze} boundary, where the inverter response becomes more sensitive to PLL synchronization error and the transition between normally evolving and frozen controller states.

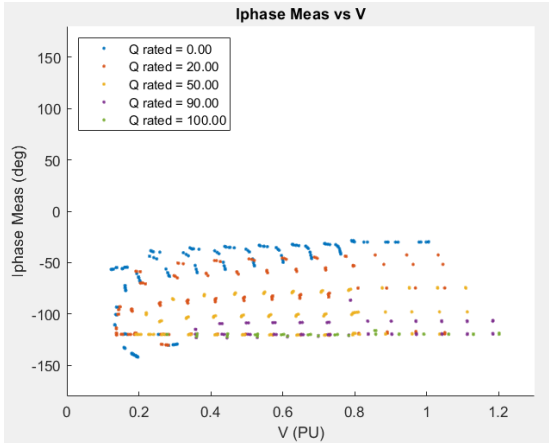


(a) Measured Current Output I_{real} at Terminal vs Terminal Voltage for Different S_{ref} .

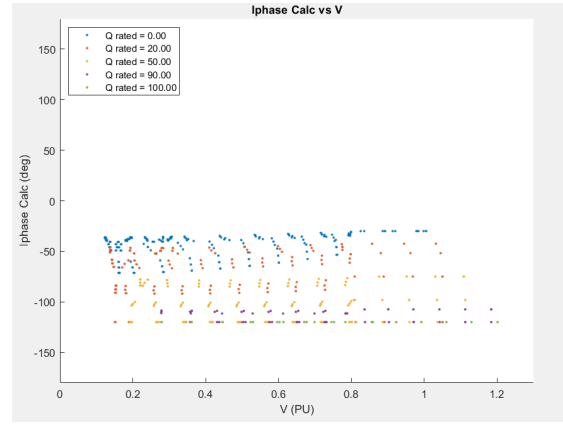


(b) Calculated Current Output I_{model} at Terminal vs Terminal Voltage for Different S_{ref} .

Figure 6.3: Comparison between the measured I magnitude at the inverter terminal in (a) and the predicted I by the model in (b).



(a) Measured Current Phase $I_{phase,real}$ at Terminal vs Terminal Voltage for Different Q_{ref} .



(b) Calculated Current Phase $I_{phase,model}$ at Terminal vs Terminal Voltage for Different Q_{ref} .

Figure 6.4: Comparison between the measured I phase at the inverter terminal in (a) and the predicted I phase by the model in (b).

The overall mean-squared errors (MSEs) are small for the three magnitude-based quantities: $\text{MSE}_P = 0.007155$, $\text{MSE}_Q = 0.002912$, and $\text{MSE}_I = 0.000964$. The particularly small current-magnitude error is significant because current is the primary output required for the proposed short-circuit application. By comparison, the current-phase error is larger, with a root-mean-squared error (RMSE) of approximately 22.06° . The results indicate that the reduced model is considerably more accurate for current magnitude than for current phase, while still retaining the main phase trends needed to represent the inverter current as a phasor instead of only as a scalar magnitude. The summary of the validation results can be found in Table 6.1.

Table 6.1: Summary of the Broader Validation Results.

Error Metrics	Short Circuit without Major Fault Phase Shift	Short Circuit with Small Fault Phase Shift ($-30^\circ, 30^\circ$)	Short Circuit with Large Fault Phase Shift ($-60^\circ, 60^\circ$)
MSE_P	0.007155	0.007741	0.008463
MSE_Q	0.002912	0.003230	0.003870
MSE_I	0.000964	0.001078	0.001568
% Failure	1.05%	1.40%	2.53%
Relaxed % Failure (error < 0.11)	0.48%	0.85%	1.93%
Relaxed % Failure (error < 0.12)	0.18%	0.24%	1.21%
RMSE θ_I	22.06°	22.17°	22.57°
Relaxed RMSE θ_I ($V > 0.3$)	8.40°	9.04°	11.53°
Relaxed RMSE θ_I ($V > 0.4$)	3.04°	3.88°	6.45°

The largest phase-model errors occur near $V \approx 0$ because the terminal-voltage phasor becomes a poor reference for the PLL. At moderate voltage, the PLL remains synchronized so that $\theta_{\text{PLL}} \approx \theta_V$, and the current phase is well approximated by the simple relationship $\theta_I \approx -\text{atan2}(Q, P) - \pi/6$. As the voltage approaches zero, the voltage vector used by the PLL becomes very small and the current orientation becomes increasingly dependent on the retained synchronization and controller states rather than only on the instantaneous Q/P ratio. This behavior is visible in Fig. 3.14, where the measured phase trajectories bend, branch, and in several cases converge toward approximately -120° below about 0.2–0.3 pu. The proposed phase model does not contain an explicit dynamic PLL state; voltage influences phase only indirectly through P_{calc} and Q_{calc} . It reproduces the dominant phase bands but cannot reproduce all of the history-dependent rotations that occur when PLL tracking becomes weak at very low voltage. The phase RMSE decreases to 3.04° when only cases with $V > 0.4$ pu are considered. This effect is even more pronounced when the fault also introduces a phase shift.

A second increase in error occurs around $V = V_{\text{freeze}} = 0.8$ pu, where the inverter changes controller operating state. The freeze logic is based on the voltage measured in the PLL-oriented d -axis, so $V_d = V \cos(\theta_V - \theta_{\text{PLL}})$. Near the threshold, even a relatively small PLL angle error can move V_d from one side of the freeze boundary to the other, causing the PI-integrator state to

continue evolving in one case but remain frozen in another. Because the model represents the low-voltage response using smooth algebraic relationships and the transition between above and below freeze state, it cannot reproduce the exact instant at which the integrator freezes or the particular stored state at that instant. This discrepancy affects P and Q as it pertains mostly to I_d and I_q , but without a large voltage phase change during the fault, the overall apparent power remains approximately the same, thus the magnitude of the current is not affected very much. It is amplified in the phase calculation because θ_I depends on their ratio through $\text{atan2}(Q, P)$; relatively small power errors near the transition can produce a noticeably larger phase error.

6.2 Two Inverters 3-Bus System

To further verify that the proposed empirical model is not limited to the single-inverter cases used during parameter identification, the model is also evaluated using the 3-bus, two-inverter network shown in Fig. 3.7. Different combinations of P_{ref} and Q_{ref} are assigned to the two inverters so that each resource operates from a different prefault condition before the fault is applied. At the same time, an analytical power-flow calculation using the empirical model is performed to find the prefault operating conditions. The resulting terminal voltages are then used to calculate P , Q , current magnitude, and current phase for each inverter after the fault, and these values are compared with the corresponding PSCAD measurements. By varying P_{ref} and Q_{ref} independently between the two resources, this test verifies whether the model preserves its dependence on prefault operating point when multiple IBRs interact through the same network rather than being evaluated in isolation.

For the example validation case presented in this section, the two inverters in the multi-bus test system were assigned different prefault operating points. The inverter connected at Bus 2 was operated with $P_{\text{ref}} = 80$ and $Q_{\text{ref}} = 20$, corresponding to 0.8 pu and 0.2 pu, whereas the inverter connected at Bus 4 was operated with $P_{\text{ref}} = 50$ and $Q_{\text{ref}} = 20$, or 0.5 pu and 0.2 pu. The two inverters have different prefault apparent-power demands, $S_{\text{ref}} \approx 0.825$ pu and 0.539 pu, respectively. This setup provides a useful verification case because the two resources experience the same interconnected network disturbance but begin from different active-power and current-loading conditions.

The proposed inverter models are embedded within a power-flow solution so that the prefault and postfault operating states can be solved consistently at the network level. The prefault power-flow calculation establishes the initial bus voltages and corresponding inverter operating conditions, while the postfault solution iteratively updates the bus voltages and uses the proposed empirical relationships to calculate the inverter active power, reactive power, current magnitude, and current phase at each bus. In this way, the IBR model is coupled directly to the surrounding network: the bus voltage determines the inverter response, while the resulting inverter current injection modifies the network voltage until a consistent solution is obtained. The final outputs include the converged V , P , Q , and I quantities at each bus for both the prefault and faulted conditions, allowing direct comparison with the corresponding PSCAD network simulation. It should be noted that the EMT results account for Q loss during the LCL filter inductor coupling. The model accounts for this by iteratively subtracting the loss generated by the calculated current from the calculated Q to find a new current until a consistent solution is obtained.

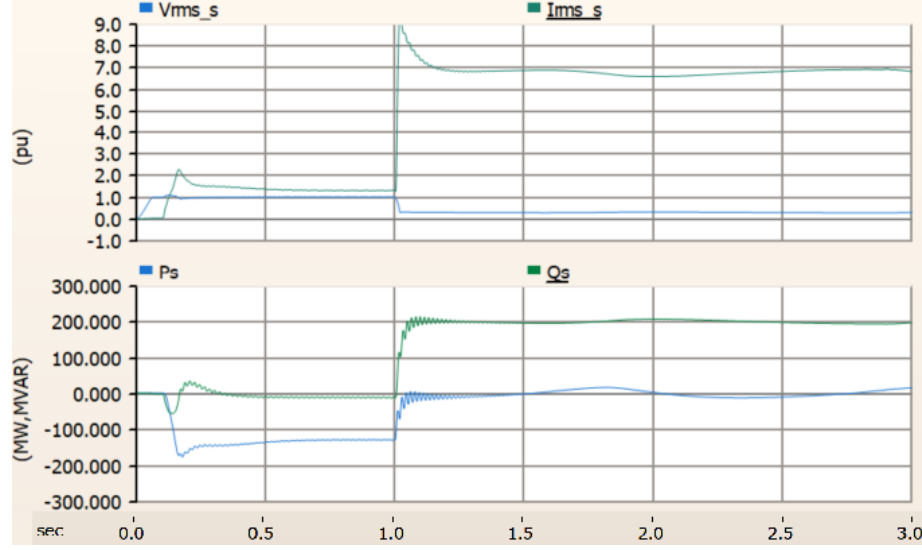


Figure 6.5: V_{rms} , I_{rms} , P , and Q Measured from the Terminals of the Synchronous Machine (Fig. 3.7) to Bus 1 with fault inception at $t = 1$ s.

The prefault EMT results in Figs. 6.5–6.8 show that the network reaches the expected steady operating condition before the fault is applied, and the analytical power-flow solution reproduces this state closely. At Bus 1, Fig. 6.5 shows the SM supplying the net power required by the two inverter operating points and the surrounding network. The analytical solution gives $V_1 = 1.000$ pu at 0° , $P_1 = -1.30$ pu, $Q_1 = -0.328$ pu, and $I_1 = 1.341$ pu, which is consistent with the steady prefault levels visible in the EMT waveform. At the fault location, Bus 3, the analytical solution gives $V_3 = 1.0067$ pu while P_3 , Q_3 , and I_3 are essentially zero before the fault. This agrees with Fig. 6.7, since the fault branch is inactive during the prefault period and carries essentially no current or power even though the bus remains energized at approximately 1 pu.

For the first inverter at Bus 2, operated with $P_{ref} = 0.8$ pu and $Q_{ref} = 0.2$ pu, the analytical power flow gives $V_2 = 1.0141$ pu, $P_2 = 0.800$ pu, $Q_2 = 0.1339$ pu, and an inverter-current magnitude of approximately 0.813 pu. Figure 6.6 shows essentially the same prefault behavior: the voltage settles close to 1 pu, the active power settles near 0.8 pu, and the current is close to the expected 0.82-pu level. Similarly, for the second inverter at Bus 4, operated at $P_{ref} = 0.5$ pu and $Q_{ref} = 0.2$ pu, the analytical solution gives $V_4 = 1.0142$ pu, $P_4 = 0.500$ pu, $Q_4 = 0.1718$ pu, and $I_4 = 0.531$ pu, again closely matching the prefault steady values in Fig. 6.8. The small difference between the commanded $Q_{ref} = 0.2$ pu and the reactive power reported at the network buses is expected because the EMT quantities in Figs. 6.6 and 6.8 are measured at the inverter-to-network terminals, after the interfacing transformer/filter and network elements.

The close correspondence between the EMT prefault waveforms and the power-flow solution establishes an important baseline for the subsequent fault comparison. Before introducing the non-linear low-voltage behavior, both approaches produce approximately the same bus voltages, active-power transfers, reactive-power flows, and inverter-current magnitudes. Differences observed after fault inception can be attributed primarily to the proposed postfault inverter representation and its treatment of voltage depression, current limiting, and controller-state effects, rather than to an inconsistency in the initial network operating point. A summary of the prefault results can be found

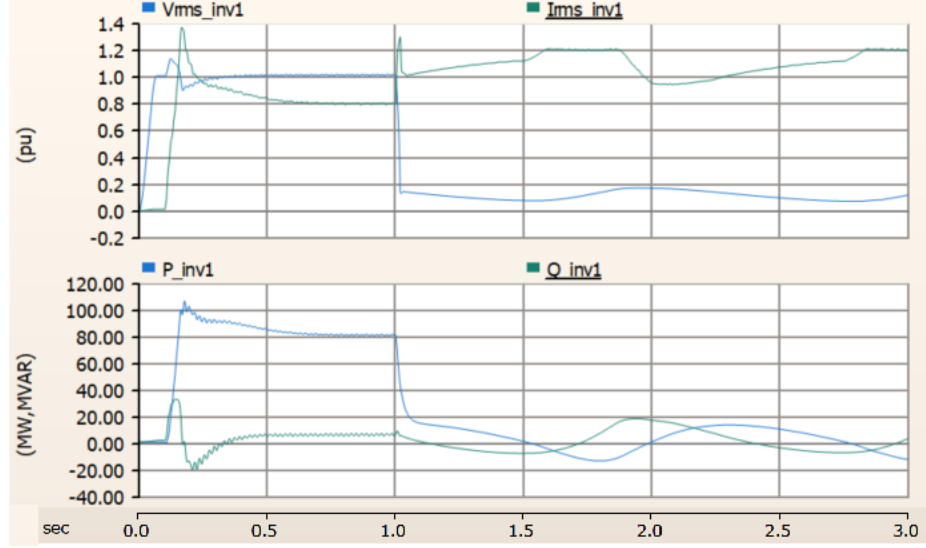


Figure 6.6: V_{rms} , I_{rms} , P , and Q Measured from the Terminals of the First Inverter (Fig. 3.7) to Bus 2 with fault inception at $t = 1$ s.

in Table 6.2.

Table 6.2: Summary of 3-Bus System prefault Results for the Inverters.

Prefault Values	Modeled Inverter 1 at Bus 2	EMT Inverter 1 at Bus 2 (Averaged)	Modeled Inverter 2 at Bus 4	EMT Inverter 2 at Bus 4 (Averaged)
V pu	1.0141	1.0146	1.0142	1.0132
P pu	0.800	0.799	0.500	0.507
Q pu	0.134	0.060	0.172	0.143
I pu	0.813	0.801	0.531	0.506

Because the EMT waveforms in Figs. 6.5–6.8 remain somewhat oscillatory after the fault, the most appropriate comparison with the analytical power-flow solution is the approximate average postfault operating level after the initial transient has decayed. The purpose of the empirical model is to represent the sustained postfault state instead of the detailed switching and controller oscillations of the EMT simulation. In the network formulation, the empirical inverter equations are embedded in the power-flow solver so that bus voltage and inverter current are updated together until a consistent postfault solution is obtained. The calculated results reproduce the main postfault trends seen in the EMT simulation: a severe voltage depression throughout the network, strong current injection from the synchronous source, saturation of the Bus 2 inverter near its 1.2-pu current limit, and a smaller current contribution from the more lightly loaded Bus 4 inverter.

At Bus 1, corresponding to the synchronous-machine terminal in Fig. 6.5, the analytical solution gives $V_1 = 0.3085$ pu and $I_1 = 6.51$ pu. The EMT waveform shows approximately the same sustained behavior, with the RMS voltage settling near 0.3 pu and the machine current increasing to roughly 6–7 pu after the fault. The analytical solution also gives $P_1 = -0.1646$ pu and $Q_1 = 2.0$ pu; from Fig. 6.5, the postfault active-power transfer remains relatively small compared with the very large

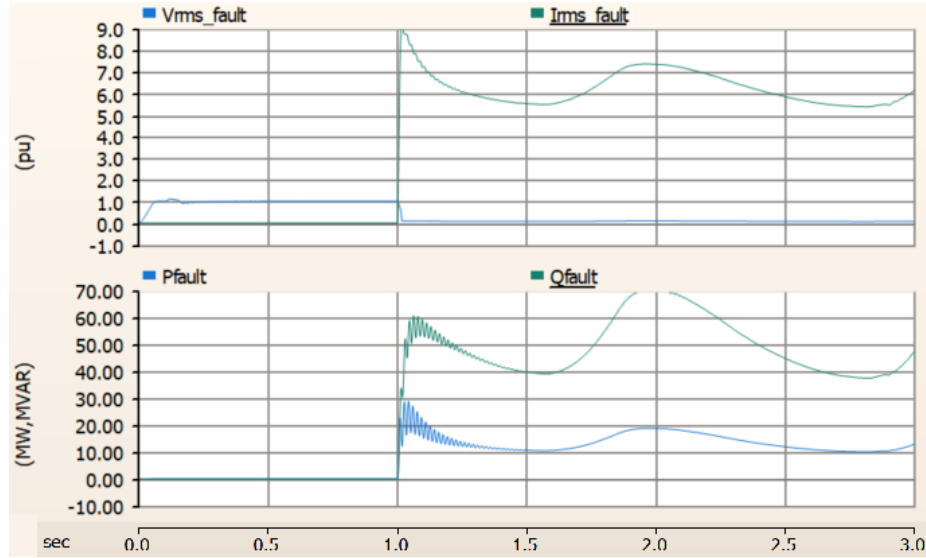


Figure 6.7: V_{rms} , I_{rms} , P , and Q Measured from Bus 3, the Fault Bus (Fig. 3.7) with fault inception at $t = 1$ s.

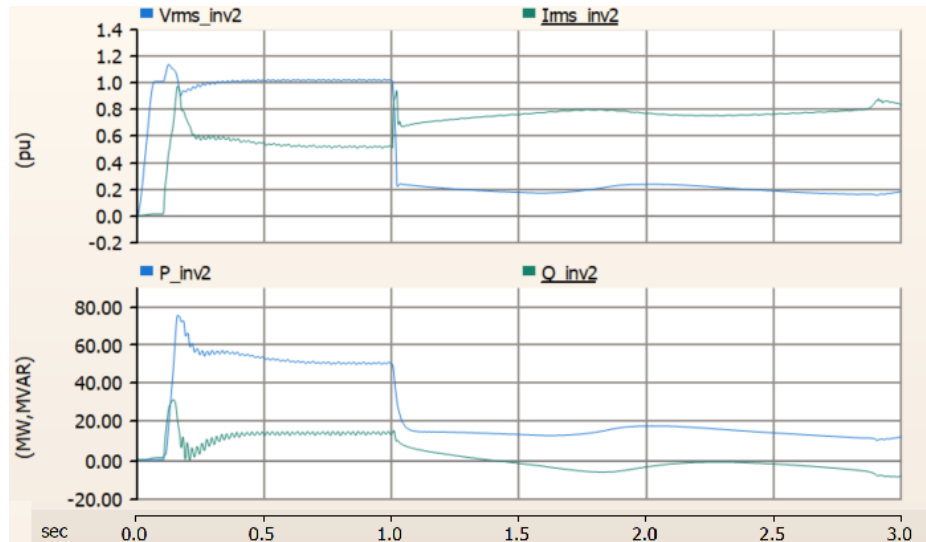


Figure 6.8: V_{rms} , I_{rms} , P , and Q Measured from the Terminals of the Second Inverter (Fig. 3.7) to Bus 4 with fault inception at $t = 1$ s.

reactive-power contribution that rises to approximately 2 pu. The close agreement is physically reasonable because, unlike the IBRs, the SM is able to supply several per-unit current during the severe voltage depression and becomes the dominant source of fault current in the network.

For the Bus 2 inverter, Fig. 6.6 shows a constrained postfault response. The proposed model predicts $V_2 = 0.1435$ pu and $I_2 = 1.20$ pu. These values agree well with the approximate EMT averages: the terminal voltage collapses to roughly 0.12 pu, while the inverter current remains close to the imposed 1.2-pu converter limit. Most importantly for the short-circuit application, the model correctly identifies that this inverter has entered the current-limited, low-voltage regime. The Bus 4 inverter provides a useful contrast because its prefault loading is smaller and it is further away from the fault location. The postfault solution using the proposed model gives $V_4 = 0.2381$ pu and $I_4 = 0.7768$ pu. Figure 6.8 shows very similar sustained levels: the inverter voltage settles at roughly 0.2–0.25 pu and the current remains around 0.75–0.8 pu. Unlike the Bus 2 inverter, Bus 4 does not reach the 1.2-pu current ceiling because its prefault apparent-power demand is lower. This is an important validation of the empirical formulation: the model does not simply force every inverter to inject $I_{\max}$ during a severe fault, but retains the dependence on the individual inverter operating point, terminal voltage, and calculated postfault P and Q .

At the faulted Bus 3, the analytical solution predicts the deepest voltage depression, $V_3 \approx 0.092$ pu, together with a network current magnitude of approximately 5.55 pu, which is consistent with the very low voltage and large fault current visible in Fig. 6.7. The near-zero P_3 and Q_3 values reported by the power-flow table should be interpreted carefully: they represent the specified/net nodal power injection at Bus 3, where no generator or load is assigned, instead of the power flowing through the fault branch itself. The EMT traces in Fig. 6.7 can show more branch power associated with the fault current even while the solver reports approximately zero net bus injection. Taken together, the postfault comparisons show that the analytical solution captures the quantities most important to the proposed short-circuit application—particularly the bus-voltage depressions and inverter-current magnitudes—while the remaining differences in P and Q are largely associated with the oscillatory controller response that a steady-state reduced-order model is not intended to reproduce exactly. The summary of the postfault results can be found in Table 6.3.

Table 6.3: Summary of 3-Bus System postfault Results for the Inverters.

Postfault Values	Modeled Inverter 1 at Bus 2	EMT Inverter 1 at Bus 2 (Averaged)	Modeled Inverter 2 at Bus 4	EMT Inverter 2 at Bus 4 (Averaged)
V pu	0.144	0.123	0.238	0.209
P pu	0.168	0.028	0.157	0.155
Q pu	-0.104	0.033	0.038	-0.030
I pu	1.200	1.116	0.777	0.767

The postfault oscillations observed in the EMT waveforms are consistent with the converter-driven interaction mechanisms discussed by Singh et al. [2] in Section 2.1.4. For a GFL inverter, the terminal voltage used by the PLL changes abruptly during the fault, causing the PLL, outer power-control loop, and inner current-control loop to readjust simultaneously while interacting with the surrounding network. After fault inception the inverter does not move instantaneously to a

new steady operating point; instead, the PLL angle, d - and q -axis current commands, and terminal voltage continue to adjust to one another, producing the low-frequency fluctuations seen in V , I , P , and Q . These oscillations are expected in the detailed EMT simulation, whereas the proposed analytical power-flow model represents the average or settled postfault operating state.

6.3 Empirical Model Convergence

To evaluate whether the proposed empirical IBR model can be incorporated into a large power-flow problem without preventing numerical convergence, a synthetic 4000-bus network was constructed in MATLAB. The network contains a connected radial backbone with additional random tie lines, conventional loads, SGs, and GFL IBRs; approximately 8% of the buses contain IBRs and 2% contain SGs. A fault was then introduced near the middle of the network by adding a shunt impedance of $Z_f = 0.02 + j0.05$ pu at approximately Bus 2000. The network is solved using a sparse Newton–Raphson power-flow formulation, with a power-flow tolerance of 10^{-8} and a maximum of 40 inner iterations.

The prefault solution is first obtained to establish V_{prefault} at every bus. For the faulted solution, the bus voltages and IBR injections are then solved iteratively: the Newton–Raphson power flow determines the network voltages for the current set of injections, after which each IBR is reevaluated using the proposed empirical model, including $V_{\text{freeze}} = 0.8$ pu, $\alpha = 0.8$, $k_Q = 0.25 + 0.175Q_{\text{ref}}$, and the 1.2-pu current limit. This outer iteration continues until the maximum change in IBR P or Q falls below 10^{-4} pu, with relaxation applied between successive solutions to improve numerical stability. The purpose of this test is not to demonstrate the physical accuracy of a particular 4000-bus system, since the network itself is synthetic, but to demonstrate that the nonlinear empirical IBR equations can be embedded in a large sparse network calculation and still reach a self-consistent postfault solution.

Chapter 7

Discussion

7.1 Sensitivity to Prefault Voltage, P , and Q

The prefault terminal voltage, V_{prefault} , has a secondary but measurable influence on the postfault inverter response because it determines the severity of the voltage depression relative to the initial operating condition. This dependence is explicitly retained in the active- and reactive-power model (5.8). For the same postfault terminal voltage V , a larger V_{prefault} produces a smaller normalized ratio V/V_{prefault} , representing a more severe relative voltage drop and a lower calculated active-power output. This dependence was introduced because the EMT results showed that cases having the same postfault voltage but different prefault voltages did not produce identical P ; normalizing by V_{prefault} allows the model to preserve this history dependence. Below V_{freeze} , the controller state is frozen and the prefault operating condition remains embedded in the subsequent response. The prefault voltage also enters the reactive-power model through the voltage-depression term $k_Q V(V_{\text{prefault}} - V)$ (5.17). For a fixed postfault voltage and Q_{ref} , increasing V_{prefault} increases the magnitude of this correction and can modify the resulting Q_{calc} . The effect then propagates into current magnitude through Equation (5.22) and into current phase through the Q/P ratio. The sensitivity to V_{prefault} is not negligible in the low-voltage frozen region where the inverter retains information about the operating state from which the fault was entered. This justifies treating V_{prefault} as an explicit model input instead of assuming that two faults producing the same final terminal voltage result in identical inverter fault-current behavior.

The ratio between P_{ref} and Q_{ref} has a strong influence on current orientation. Since P primarily determines I_d and Q determines I_q , the current phase is approximately governed by $\text{atan2}(-Q, P)$. Two operating points can have the same S_{ref} , and hence similar current magnitudes, while producing different current phases because their $Q_{\text{ref}}/P_{\text{ref}}$ ratios are different. This behavior is consistent with Fig. 3.14, where increasing the reactive-power reference rotates the current toward increasingly negative phase angles. From a protection perspective, this distinction is important because two IBRs supplying nearly identical fault-current magnitudes may produce different current angles, active/reactive powers, and apparent impedance. Below $V_{\text{freeze}} = 0.8$ pu, the references become even less like instantaneous commands: they continue to influence the response by determining the prefault I_d/I_q operating point and controller state entering the fault.

7.2 Sensitivity to Current Limiter and Freeze State Voltage Threshold

The converter current limit, $I_{\max}$, directly controls the maximum sustained fault-current magnitude and has a strong influence once the unconstrained current demand approaches the converter capability. Once this boundary is crossed, the current becomes approximately fixed at $I_{\max}$. Reducing $I_{\max}$ causes the inverter to enter saturation at a higher terminal voltage and lowers the maximum fault-current contribution. Increasing $I_{\max}$ permits the inverter to retain its P/Q objectives over a wider voltage range before saturation occurs. In the EPRI model studied here, $I_{\max} = 1.2$ pu and reactive-current priority is used, so saturation also affects the allocation between P and Q : reactive current is retained preferentially while the remaining capability is available for active current. The current limit changes the feasible $P - Q$ operating region and the governing relationship between voltage, power, and fault current.

The low-voltage state-freeze threshold, V_{freeze} , determines when the inverter changes control regime. Above this threshold, the relevant PI-integrator states continue to evolve and the controller can adjust I_d and I_q toward the instantaneous P_{ref} and Q_{ref} requirements. Once V_d falls below the threshold, the relevant integrator action is disabled and the subsequent response becomes dependent on the controller state at the instant of freezing. Increasing V_{freeze} would cause the inverter to enter the frozen-state model for a less severe voltage depression, extending the range over which the empirical low-voltage P and Q relationships apply; decreasing V_{freeze} would allow normal closed-loop regulation to continue to a lower voltage. This parameter can also have a particularly strong effect on current phase because the freeze decision is based on $V_d = V \cos(\theta_V - \theta_{\text{PLL}})$. Near the threshold, a relatively small voltage-angle or PLL error can cause otherwise similar cases to fall on opposite sides of the freeze boundary and subsequently develop different orientations.

7.3 Sensitivity to Parameters α and k_Q

The parameter α controls the rate at which active power decreases with voltage below the freeze threshold through Equation (5.8). Increasing α causes P_{calc} to decrease more rapidly as voltage collapses, while decreasing α allows the inverter to retain more active power at the same postfault voltage. The fitted value $\alpha = 0.8$ lies between the ideal constant-power limit ($\alpha = 0$) and the approximately frozen- I_d relationship ($\alpha = 1$), matching the EMT simulation observations. The influence of α propagates directly into current magnitude through Equation (5.22) and into current phase through the Q/P ratio. Its sensitivity is greatest in the low-voltage unsaturated region. Once the calculated current reaches $I_{\max}$, further changes in α have much less influence on current magnitude because the current is already constrained by the converter limit.

The parameter k_Q controls the magnitude of the additional nonlinear reactive-power correction, Equation (5.17). A larger k_Q increases the reactive-power contribution associated with the voltage depression, while a smaller k_Q makes the response approach the simpler frozen-current baseline $Q \approx (V/V_{\text{ref}})Q_{\text{ref}}$. The sensitivity is not uniform with voltage because the correction contains the factor $V(V_{\text{prefault}} - V)$: it becomes small as $V \rightarrow 0$ and vanishes when $V = V_{\text{prefault}}$. The model also makes k_Q explicitly dependent on Q_{ref} .

Together, α and k_Q determine how the empirical model departs from the idealized constant-

power relationships after the inverter enters its low-voltage operating regime. α primarily changes the active-power trajectory, whereas k_Q primarily changes the reactive-power trajectory; both ultimately influence current magnitude and current phase. These parameters should be interpreted as identified empirical coefficients: changing them modifies the shape of the low-voltage response, while the current limiter and V_{freeze} still impose the dominant physical boundaries on the final fault-current behavior.

Although the present thesis adopts a constant value of α and a linear relationship for k_Q , it does not imply that the coefficients must follow the same structure for every inverter implementation. They represent the lowest-order empirical relationships that provided sufficient accuracy for the EMT model dataset examined here. If measurements from another inverter reveal stronger nonlinear dependence on operating points, α and k_Q could instead be expressed using higher-order polynomials, piecewise functions, or more general nonlinear relationships and fitted to the measured inverter response using the same optimization framework. The appropriate model complexity depends on both the characteristics of the available empirical data and the accuracy required by the system operator. A relatively simple parameterization may be preferable for rapid large-system short-circuit studies, whereas applications requiring greater precision could justify additional fitted coefficients or more flexible nonlinear functions, provided that the resulting improvement in accuracy outweighs the added computational and modeling complexity.

7.4 Physical Significance of Current Phase

The current-phase results provide a physical indication of how the inverter allocates its limited current between active- and reactive-power production. For a GFL inverter whose PLL remains approximately synchronized with the terminal voltage, the d -axis current is primarily associated with active power while the q -axis current is associated with reactive power. At intermediate and higher voltages this relationship remains relatively well behaved because the PLL can continue tracking the terminal-voltage angle. As the voltage approaches V_{freeze} , and especially as $V \rightarrow 0$, the phase trajectories become increasingly nonlinear because the current orientation is not determined only by the instantaneous Q/P ratio. PLL synchronization error and the controller state retained when the low-voltage freeze condition is entered can also influence the orientation of the current vector. The error observed in current phase reflects information associated with PLL and controller dynamics that is intentionally reduced in the algebraic formulation.

7.5 Origin of Nonlinear Loops and Multiple Trajectories

The loops, branching, and scatter observed in the EMT results arise because the inverter response is not determined solely by the instantaneous terminal voltage. Once the low-voltage state-freeze condition is entered, the relevant PI-integrator states do not evolve normally. The resulting fault current depends on the present values of V , P_{ref} , and Q_{ref} and on the internal controller state that existed when the threshold was crossed. This introduces a form of history dependence: two simulation cases can reach approximately the same terminal-voltage magnitude yet produce different P , Q , or current phase because they arrived at that voltage from different operating conditions or controller trajectories. When the voltage subsequently changes, the retained state can cause the

response to follow a different path from the one followed during the initial voltage collapse, producing the loop-like or multiple-valued trajectories seen in the EMT results.

The branching becomes particularly noticeable near V_{freeze} because a relatively small difference in voltage angle or PLL tracking error can place two otherwise similar cases on opposite sides of the freeze boundary, after which their controller states evolve differently. At very low voltage, the terminal-voltage vector becomes a weak synchronization reference, increasing the influence of retained controller and synchronization states on the current orientation. At the same time, current limitation and reactive-current priority constrain the allowable $I_d - I_q$ vector, adding another nonlinear boundary to the response. The observed loops and scatter should be interpreted as the combined effect of controller-state memory, threshold switching, PLL tracking, current limiting, and current-priority logic, instead of random numerical variations. They are a physical indication that the detailed EMT response is path-dependent, which also explains why a simple single-valued constant-current or constant-power relationship cannot reproduce every trajectory exactly.

7.6 Why the Reduced Analytical Model Works

The proposed model allows for rapid calculation because it does not attempt to reproduce every internal state and transient of the detailed EMT controller. Instead, it preserves the dominant physical relationships that determine the sustained fault response. At the fundamental level, the model retains the GFL relationship between voltage, active and reactive power, and the dq -axis currents, so that P primarily determines I_d , Q primarily determines I_q , and their vector combination determines the resulting current magnitude and orientation. The nonlinear behavior observed in the EMT simulation results is then represented only where the ideal relationships cease to be sufficient: the dependence on prefault operating condition and voltage depression is retained through the empirical P_{calc} and Q_{calc} equations, while the low-voltage regime is separated from normal operation by V_{freeze} . In this way, the model captures the principal changes in operating regime without requiring explicit differential equations for the PLL, PI integrators, current controllers, filters, and other internal states.

A particularly important reason for the model's accuracy is that the fault-current magnitude is not fitted independently as an arbitrary empirical curve. Instead, once P_{calc} and Q_{calc} have been determined, the current is reconstructed using the fundamental power-current relationship and then constrained by the physical converter limit as in Equation (5.22). This preserves consistency among P , Q , and I , while reproducing the experimentally observed transition from a voltage- and operating-point-dependent current to a sustained current-limited response. The approach succeeds because the empirical portion of the model is concentrated on the nonlinear behavior that cannot be represented adequately by a conventional constant-power model, while well-established physical relationships are retained wherever possible. The result is a reduced-order representation that captures the dominant effects of terminal voltage, P_{ref} , Q_{ref} , prefault voltage, state freezing, reactive-current behavior, and I_{max} , without carrying the computational burden of the full EMT implementation.

This reduction also explains both the strength and the limitations of the model. It is well suited to estimating the average or sustained postfault operating point. It is not intended to reproduce the detailed oscillations, loops, or controller trajectories generated by the internal dynamic states of the EMT model. The fact that a relatively compact set of algebraic relationships can reproduce

the principal fault-current behavior supports the central contribution of this thesis: for rapid short-circuit calculations, it is not necessary to reproduce every internal inverter state if the reduced model retains the physical mechanisms that most strongly determine the electrical quantities of interest.

7.7 Limitations of the Proposed Framework

The proposed framework is intentionally a reduced-order representation and does not reproduce all of the transient behavior contained in the detailed EMT model. This distinction is particularly important during the first portion of the fault, because the inverter filter and severity of the voltage disturbance can strongly influence the initial current transient, while current limiting becomes more dominant in the later sustained response. The model is best interpreted as a tool for rapid estimation of the controlled or averaged postfault current.

A second limitation is the representation of current phase and internal controller-state behavior. Although current magnitude can be reconstructed accurately from the modeled P and Q together with the physical current limit, current phase depends additionally on PLL synchronization and the controller state retained during severe low-voltage conditions. At very low voltages, selected controller states are not continuously updated, introducing history dependence that cannot be represented completely by the limited algebraic equation. This is reflected in the larger error observed for current phase compared with current magnitude and in the loops and branching present in the PSCAD data. The model captures the dominant orientation associated with the P/Q allocation, but it cannot reproduce every PLL-dependent rotation or trajectory generated by the full dynamic controller.

Finally, the identified coefficients and operating boundaries are specific to the inverter model and dataset examined in this thesis. The EPRI GFL model considered here uses reactive-current priority, $I_{\max} = 1.2$ pu, and $V_{\text{freeze}} = 0.8$ pu, and the empirical values $\alpha = 0.8$ and k_Q were obtained from the corresponding EMT model response rather than derived as a universal converter constant. Other inverter manufacturers or controller implementations may use different current-priority schemes, PLL settings, freeze thresholds, or low-voltage control strategies and would require their own parameter identification. In addition, the present framework is developed primarily for balanced three-phase faults, where negative- and zero-sequence quantities can be neglected; extending the method to asymmetrical faults would require additional sequence-dependent modeling and current-allocation logic.

7.8 Engineering Interpretation and Overall Implications

From an engineering perspective, the principal implication is that much of this complicated EMT behavior can be represented with a small number of physically meaningful algebraic relationships. By retaining the dominant dependencies, the proposed framework provides a practical compromise between detailed EMT simulation and overly simplified short-circuit representations. The model remains tied to physical quantities that are meaningful to system operators such as terminal voltage, prefault voltage, P_{ref} , Q_{ref} , $I_{\max}$, and V_{freeze} —while empirical coefficients such as α and k_Q allow measured inverter behavior to be incorporated into the reduced formulation. The successful embedding of the model in the multi-inverter network and the 4000-bus sparse power-flow test further

indicates that this type of representation can be used in large-system calculations without requiring the computational complexity of a full EMT model for every individual resource.

The broader implication is that EMT models can be complemented by a faster empirical representation for applications where the sustained fault-current magnitude and approximate phasor response are the quantities of primary interest. Detailed EMT simulations remain necessary for studying initial transient peaks, oscillations, PLL dynamics, and manufacturer-specific control interactions, while the proposed framework is better suited to repeated short-circuit, planning, and protection studies in which computational speed and scalability are important. The results support the central premise of this thesis: if the dominant controller boundaries and physical power-current relationships are preserved, a reduced-order model fit to pre- and postfault measurements can reproduce the most important characteristics of IBR fault contribution while remaining simple enough for large-network implementation.

Chapter 8

Conclusions & Future Work

8.1 Conclusions

The proposed framework was shown to reproduce the quantities most important to rapid short-circuit calculations while avoiding the computational complexity of the complete EMT model. Using a 0.1-pu current-magnitude error criterion, 98.95% of cases in the overall dataset are predicted within 0.1 pu. The remaining larger errors were concentrated mainly at extreme P_{ref} - or Q_{ref} -dominated operating points. The relatively small current error is particularly significant because current magnitude is the principal output required for the intended short-circuit application. The current-phase model reproduced the dominant P/Q -dependent phase trends but had a larger RMSE of approximately 22.06° , reflecting the difficulty of representing PLL and frozen-controller history with a purely algebraic model. Network-level validation further showed that the proposed model reproduces the approximate sustained postfault behavior of a multi-inverter EMT system, including current saturation and differences caused by inverter loading, while the 4000-bus convergence study demonstrated that the nonlinear empirical model can be embedded successfully within a large sparse power-flow calculation. Overall, the results support the central conclusion that a compact, physics-informed empirical model can provide an effective middle ground between overly simplified conventional short-circuit representations and computationally intensive EMT simulation for estimating sustained IBR fault contributions.

The primary contribution of this thesis is the development of a physics-informed empirical framework for rapid calculation of the sustained short-circuit response of GFL IBRs. The proposed formulation preserves the dominant nonlinear dependencies identified from PSCAD EMT simulations. These include terminal-voltage magnitude, prefault voltage, P_{ref} , Q_{ref} , converter current limitation, reactive-current priority, and the low-voltage state-freeze boundary. Separate empirical relationships were developed for postfault active and reactive power, after which current magnitude was reconstructed from the physical complex-power relationship and constrained by I_{max} . A corresponding current-phase formulation was also developed to retain the dominant P/Q -dependent current orientation, allowing the inverter to be represented as a phasor.

A second contribution is the demonstration that this reduced representation can retain useful accuracy while remaining suitable for network-level calculations. The empirical coefficients are formulated so that they can be reidentified from measured inverter data and, if necessary, replaced

by higher-order or nonlinear functions when greater precision is required. The resulting model was validated against PSCAD data, extended from the original single-inverter cases to an interconnected two-inverter network, and embedded within an iterative power-flow formulation in which inverter injections and network voltages are solved self-consistently. Finally, successful convergence of the framework in a synthetic 4000-bus faulted network demonstrates its potential scalability for large-system studies.

The principal reduction in accuracy occurs in the current-phase prediction, for which an RMSE of approximately 22.06° was obtained. The phase model successfully reproduces the dominant P/Q -dependent phase bands, but it cannot fully reproduce the bending, branching, and history-dependent trajectories observed near V_{freeze} and at very low terminal voltage because the reduced formulation does not explicitly solve the dynamic PLL and frozen controller states. The network-level comparison nevertheless showed good agreement between the solution derived from the proposed model and the approximate sustained EMT response, particularly for bus-voltage depression and inverter-current magnitude. Overall, the accuracy results indicate that the proposed model is more reliable for sustained current magnitude than detailed current phase, which is consistent with its intended role as a computationally efficient short-circuit model.

The engineering significance of this work is that it provides a practical bridge between detailed EMT simulation and overly simplified conventional short-circuit models. By retaining the dominant physical dependencies of the sustained IBR fault response, the proposed framework can estimate fault-current behavior with lower computational complexity while remaining suitable for network-level studies. This makes the approach relevant to transmission planning and protection applications where many operating conditions, contingencies, or large networks must be evaluated and a full EMT representation of every inverter would be impractical. Overall, the thesis shows that system operators do not necessarily need access to every proprietary internal controller state to obtain a useful estimate of sustained IBR short-circuit contribution; instead, appropriately identified empirical relationships can preserve the most important external behavior while offering the speed and scalability required for large-system engineering studies.

8.2 Future Work

One important extension of this work would be to apply the same empirical modeling philosophy to GFM inverters. The present framework is developed specifically around a GFL inverter whose behavior is organized around PLL synchronization, d - and q -axis current control, reactive-current priority, and low-voltage state freezing. GFM inverters instead regulate voltage and frequency and are more naturally represented as controlled voltage sources, so their fault response and current-limiting transitions differ from those of GFL resources. Future work could identify an equivalent set of low-order empirical relationships for GFM resources, including the effects of droop control, virtual impedance, voltage regulation, and the transition from normal voltage-forming operation into current-limited operation. Comparing GFL and GFM empirical models within the same network would also help determine whether a common reduced framework can be developed for mixed inverter-dominated systems.

A second major extension is the treatment of unsymmetrical faults and sequence-domain behavior. The present thesis focuses primarily on balanced three-phase faults, for which negative- and zero-

sequence quantities can be neglected. Single-line-to-ground, line-to-line, and double-line-to-ground faults would require explicit representation of positive-, negative-, and potentially zero-sequence quantities. For IBRs, this is especially important because negative-sequence current is determined by the implemented control strategy rather than by a fixed machine sequence impedance, and positive- and negative-sequence currents can interact through the physical phase-current limit. A future sequence-domain empirical model could estimate both I_1 and I_2 , including their magnitudes, relative phase angles, current-priority rules, and the resulting maximum phase current. Such an extension would make the framework more directly applicable to protection functions that depend on negative-sequence quantities.

Further work should also examine different PLL and controller implementations and more advanced parameter-identification approaches. The phase results in this thesis show that PLL behavior becomes particularly important near V_{freeze} and at very low voltage, while Section 2 also indicates that PLL and current-loop dynamics can become strongly coupled during severe disturbances. The current coefficients α and k_Q were intentionally kept simple, but future models could identify these relationships from larger datasets using nonlinear regression, higher-order polynomial fitting, Gaussian processes, neural networks, or other machine-learning techniques. The objective would be to use machine learning to determine the empirical correction terms or controller-dependent parameters while preserving physical constraints such as $I \leq I_{\text{max}}$. This would allow the required model complexity to adapt to the measured behavior and accuracy requirements of different inverter technologies.

Finally, the framework should be validated using real inverter hardware and field measurements instead of only a generic PSCAD EMT model. The present coefficients and control boundaries are specific to the EPRI GFL model used in this study, and other manufacturers may implement different PLLs, current-priority schemes, freeze thresholds, anti-windup functions, and proprietary fault-ride-through controls. Hardware-in-the-loop testing, laboratory converter measurements, or disturbance recordings from operating renewable plants could be used to obtain terminal V , P , Q , and current-phasor data over a controlled range of faults. The empirical parameters could then be identified directly from measured external behavior without requiring access to proprietary internal controller details. Such validation would be an important step toward determining whether the proposed approach can serve as a practical utility-oriented method for constructing fast, vendor-specific or vendor-agnostic short-circuit models from measured inverter responses.

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